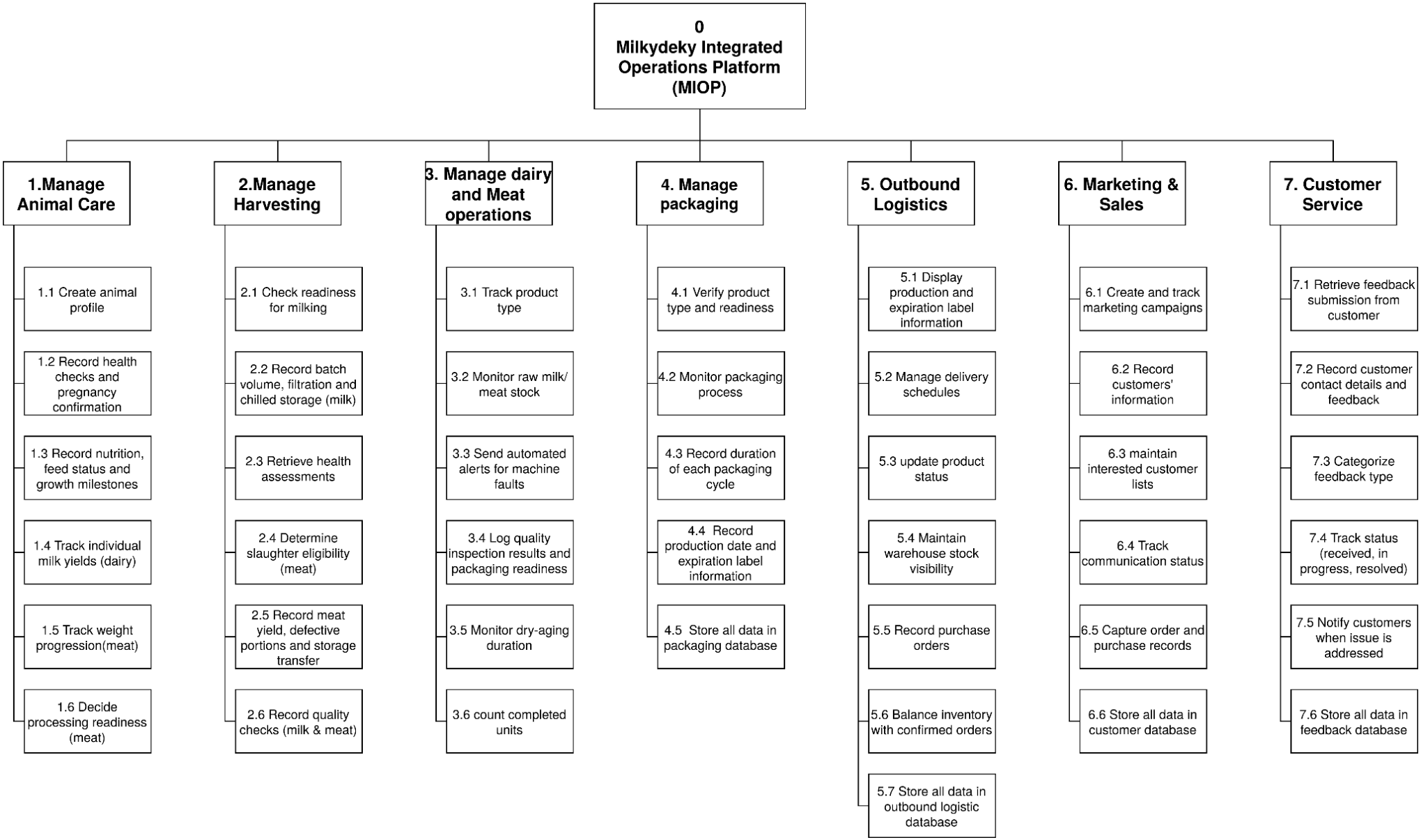
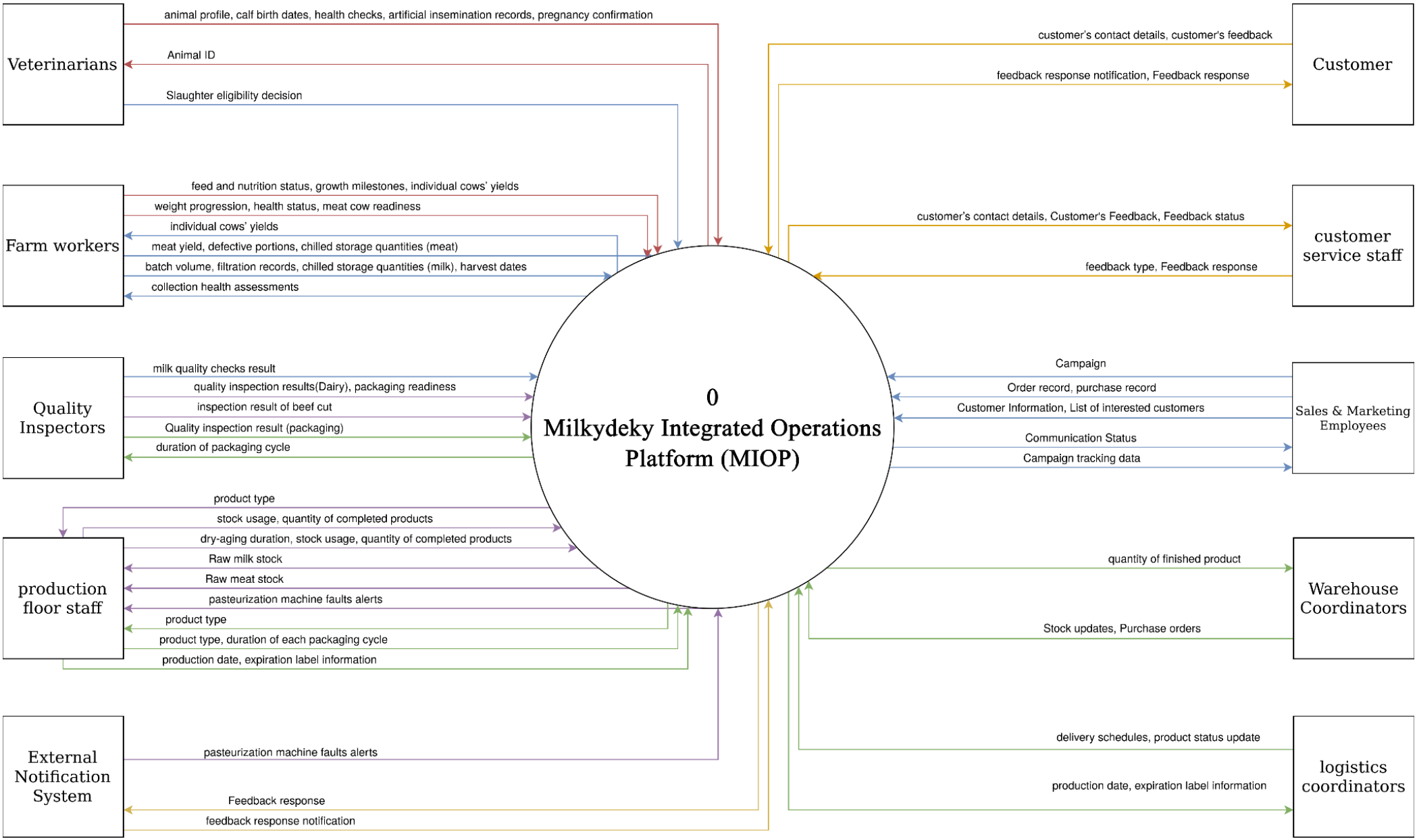
Detailed Analysis & Design Diagrams

Functional decomposition diagram (Structure chart)

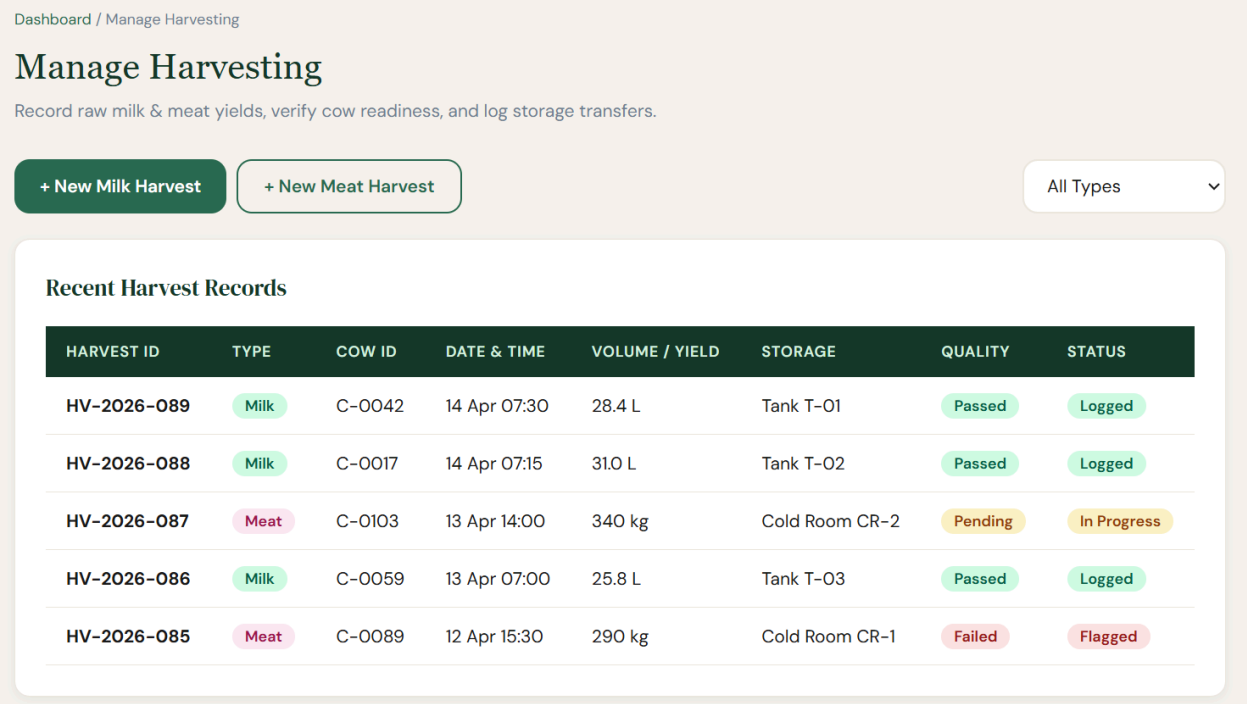


Data Flow Diagrams
Level 0



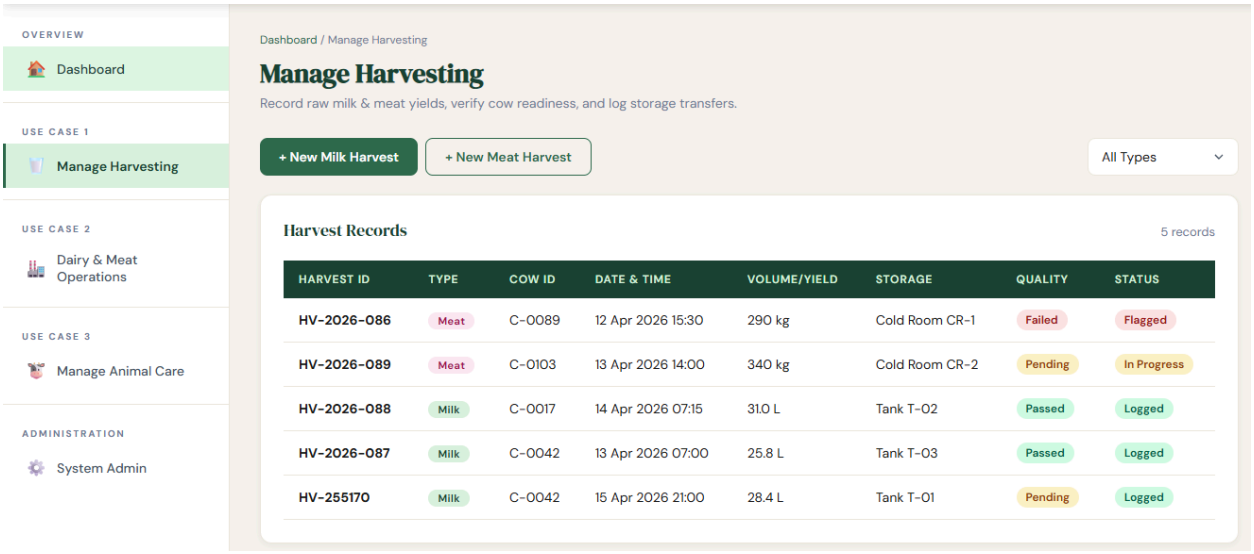
Step 2: When clicking the “Open Module” button, the system will go to the page of the desired process.

Example: When clicking the “Open Module” button of the **Manage Harvesting** process.



Manage Harvesting

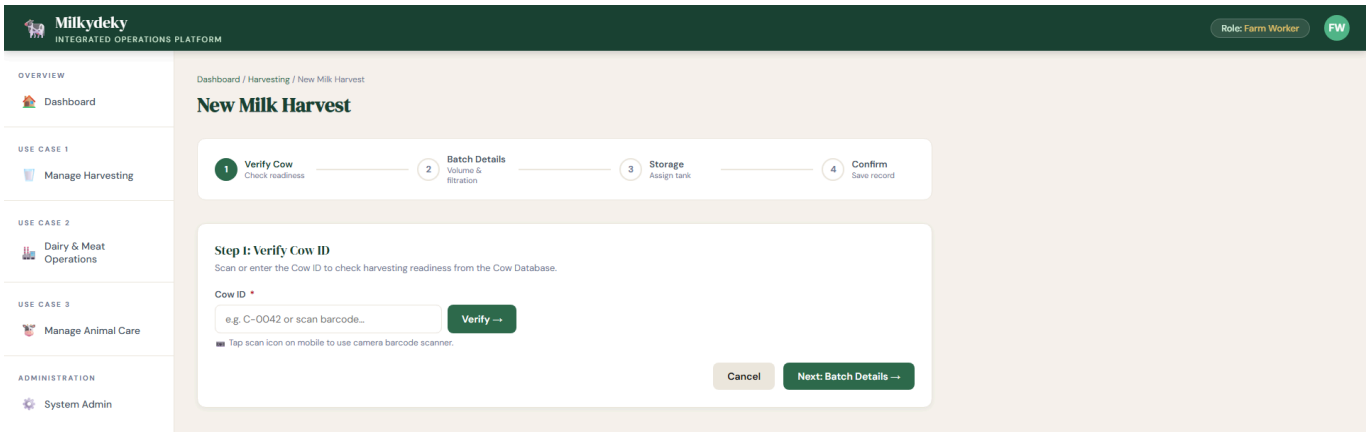
This page shows the management of harvesting operations. Farm workers can perform two primary operations 1.Adding a new milk harvest and 2.Adding a new meat harvest. The page also features the “Recent Harvest Records” for checking the harvest information of cows after harvest such as volume/yield, storage location, quality and status for specific cows. Our system has a function type (All Types, Milk Only, Meat Only) for the selection field to make it easier for farmworkers to view information for each raw material.



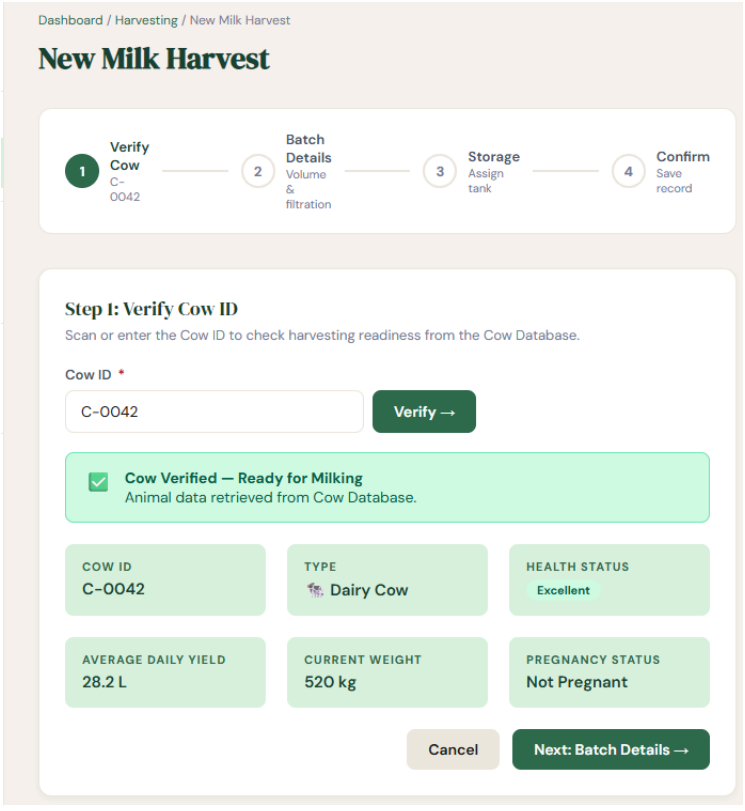
Add New Milk Harvest:

After clicking on "+New Milk Harvest". The farmworker is directed to the New Milk Harvest page. This page features the step box. It has 4 steps 1. Verify Cow 2. Batch Details 3. Storage 4. Confirm.

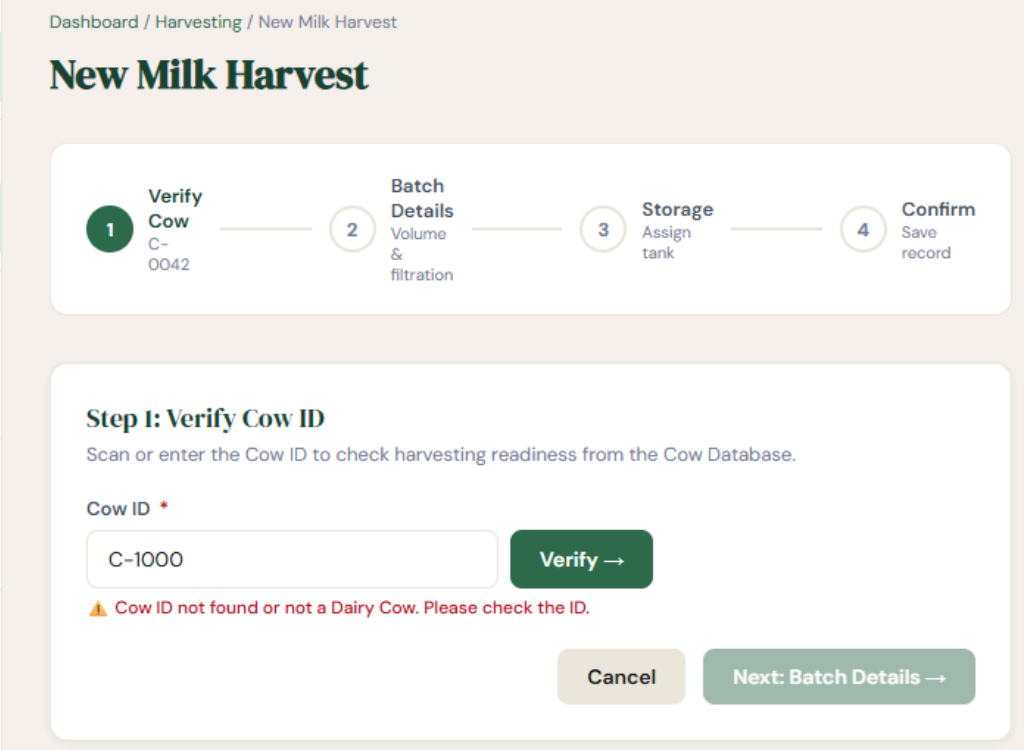
Step1: Verify Cow ID, the farm worker needs to enter the Cow ID to verify and click "Next: Batch Details" to go to the next step.



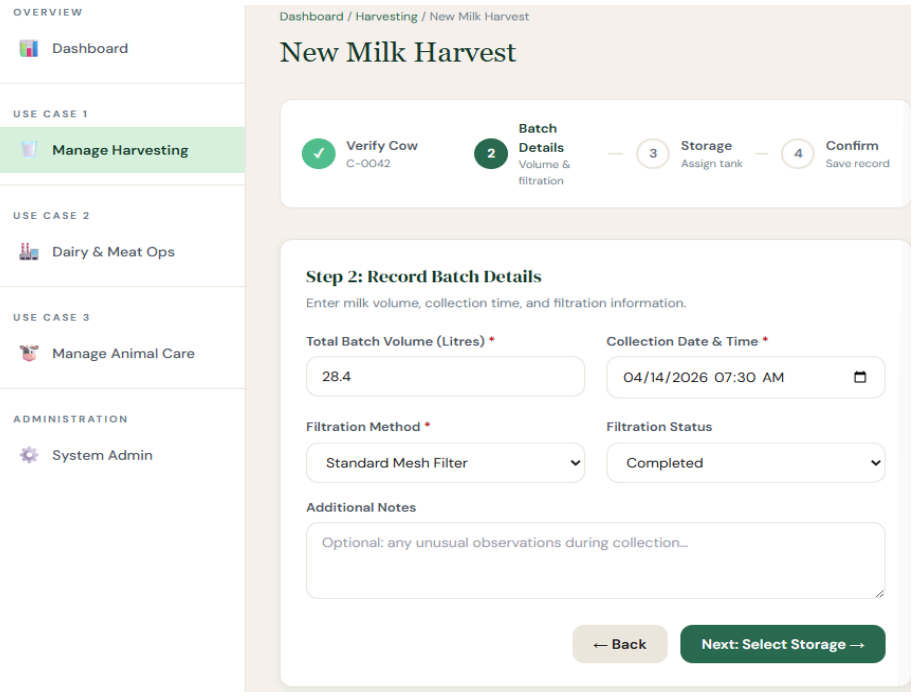
After the farm worker enters the cow ID. On that page will show the Cow verified. And will show the details of that cow such as 1.Cow ID, 2.Type of cow, 3.Health status, 4. Last milked(Last harvest of that cow), 5. Average daily yield, and 6. Pregnancy status.



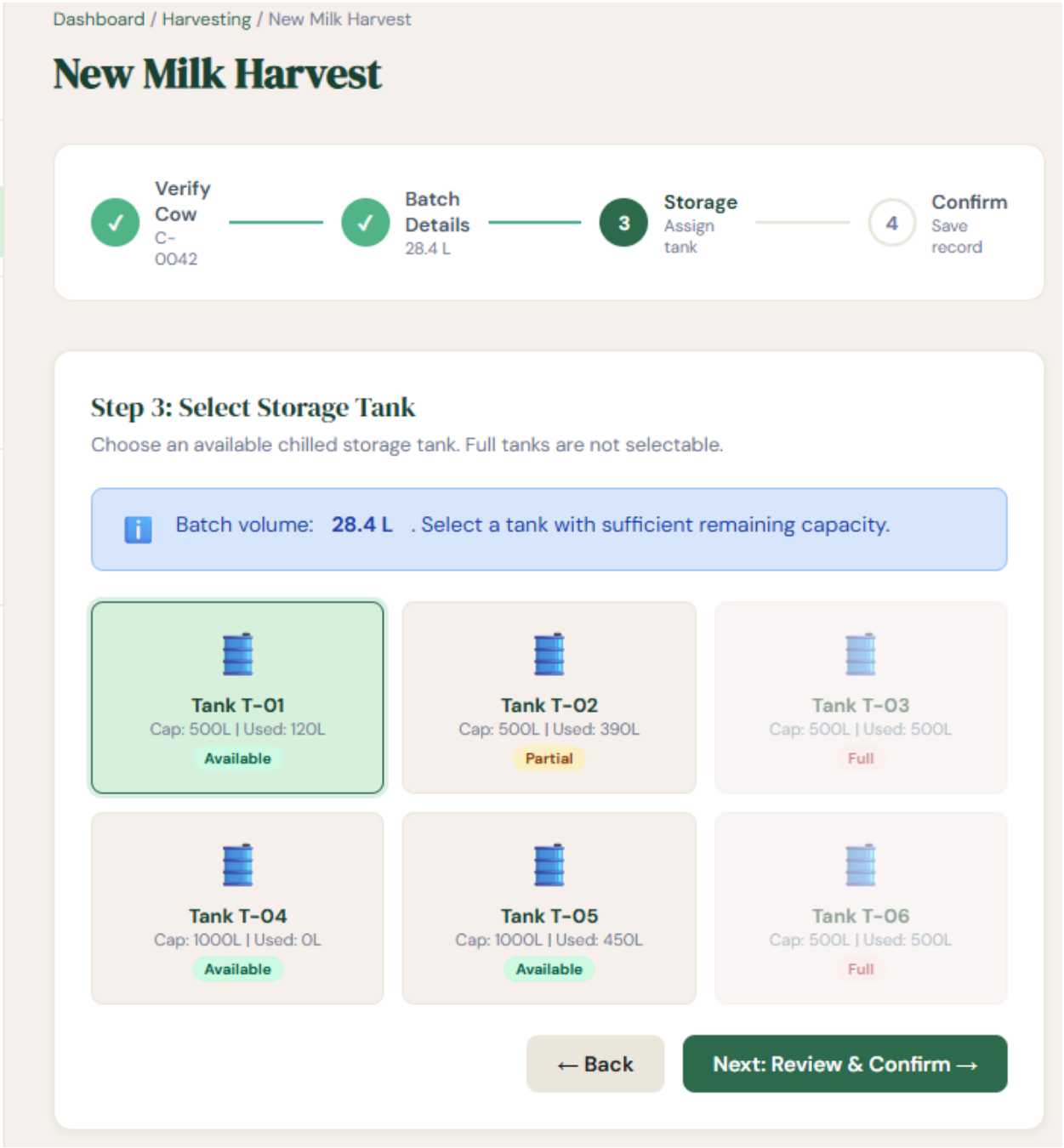
If the farm worker puts the wrong cow ID it will show the " Cow ID not found or not ready for milking. Please check the ID." and can't go to the next page.



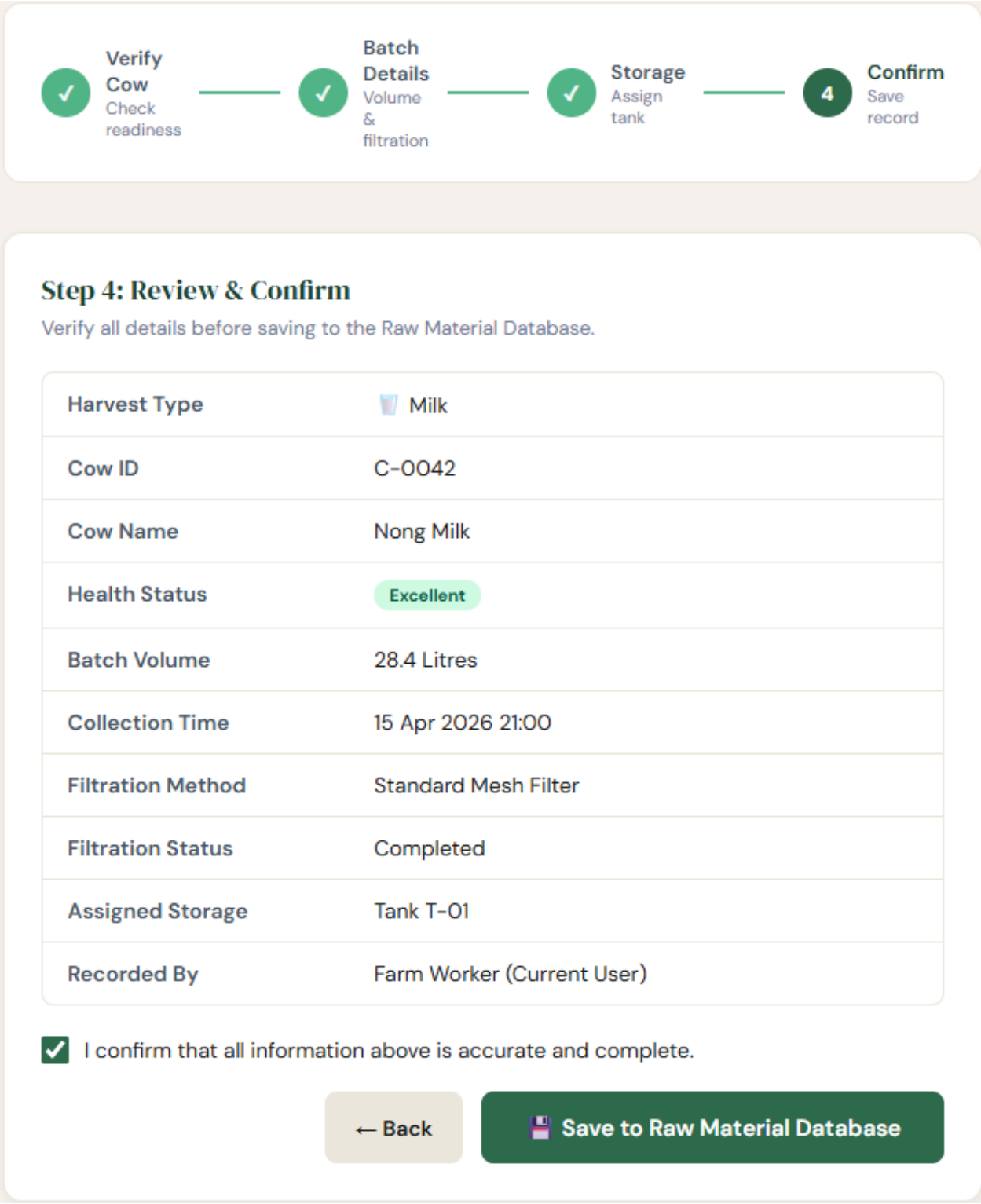
Step2: Record Batch Details, This page will have the employees enter the information about 1. Milk Volume, 2. Collection date&time, 3. Filtration Method (Standard Mesh Filter, Membrane Filtration, Centrifugal Separation), 4. Filtration Status(Completed, Pending, Skipped (Emergency)), and 5. Additional Notes for some observation during collection.



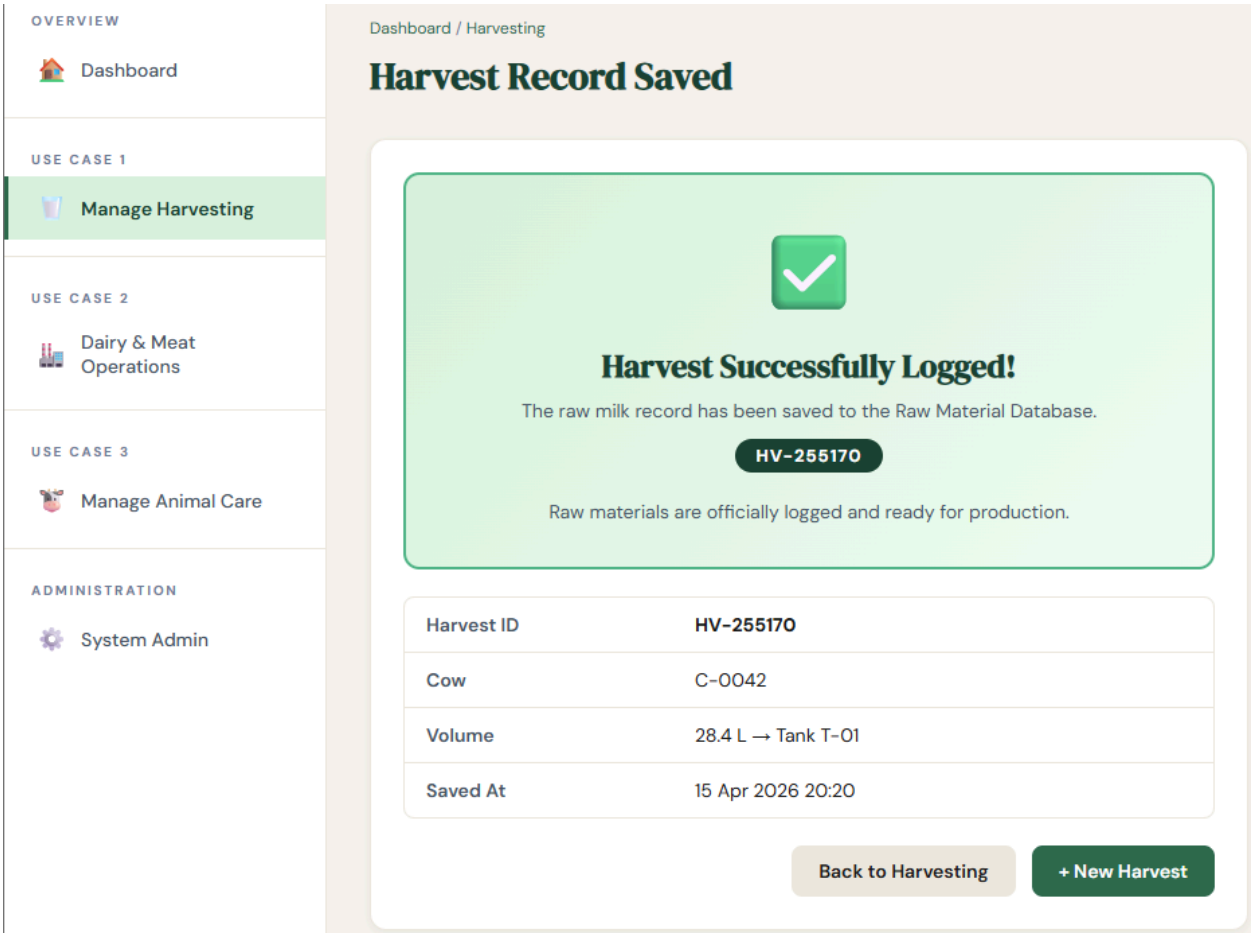
Step3: Farm workers assign the newly harvested milk to a specific chilled storage tank. It has a blue information banner to display the current milk volume that farmworkers need to put in the tank and display all tanks containing raw cow milk and farmworkers can select the desired tank to store fresh cow milk. If a tank still has space to hold raw material, it will be highlighted in green as “Available”. If a tank is more than half full, it will be indicated in yellow as "Partial." If a tank is full, it will be indicated in red as "Full" and can no longer be used.



Step4: Review & Confirm. This is the last page of add new milk harvest. It will show all details that farmworkers add to the system and after the farmworker checks all the information, farmworkers need to click “Save Harvest Record” to save all information.



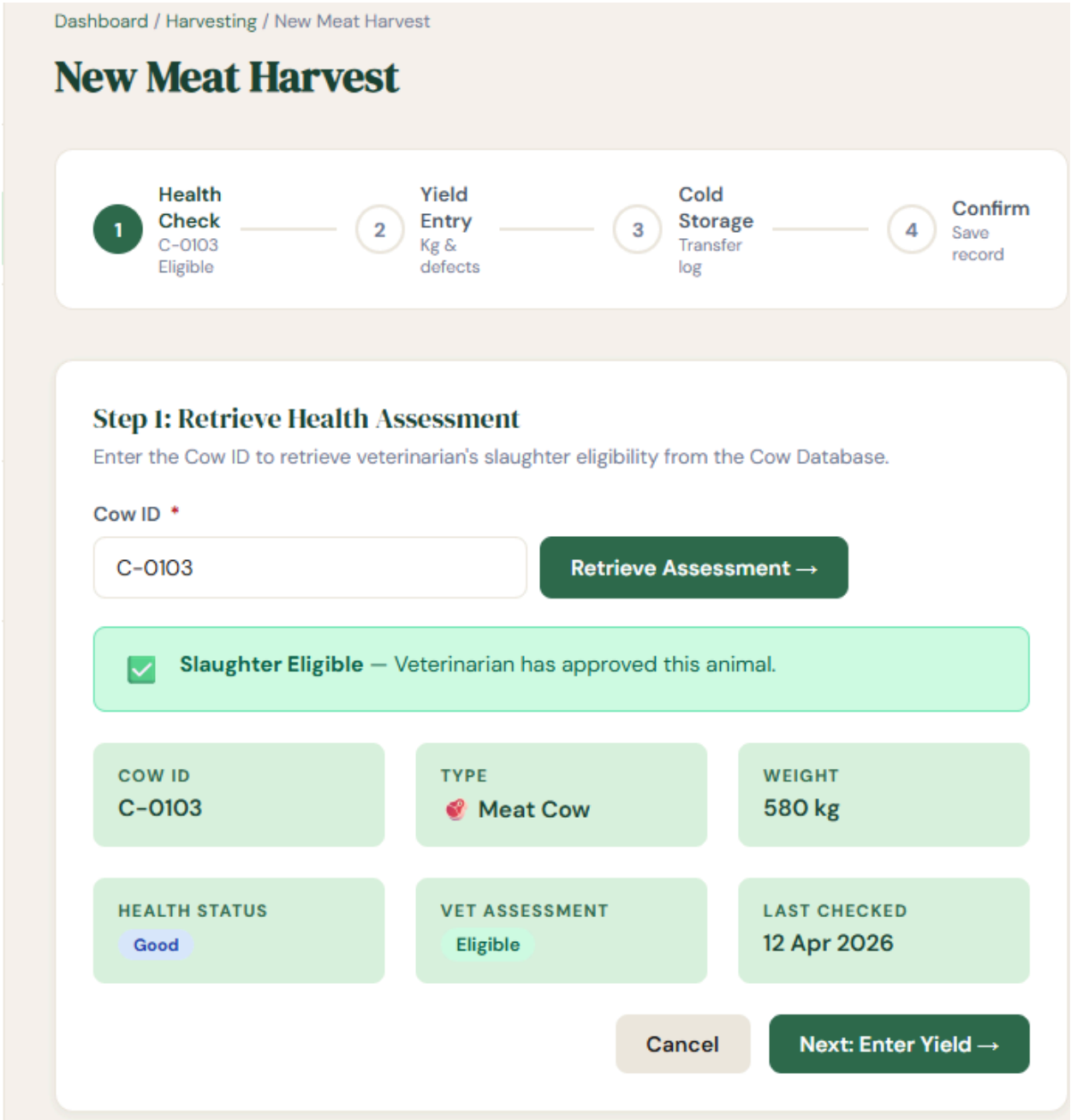
Finish the record new milk harvest. And it will show the 1.Cow ID, 2.Volume of milk to which tank, 3. Saved At date&time. And have 2 buttons for next to back to harvesting or add new milk harvest.



Add New Meat Harvest

After clicking on "+New Meat Harvest". It will go to the Meat Milk Harvest page, and it will show the step box. It has 4 steps 1. Health Check, 2. Yield Entry, 3. Cold Storage, 4. Confirm.

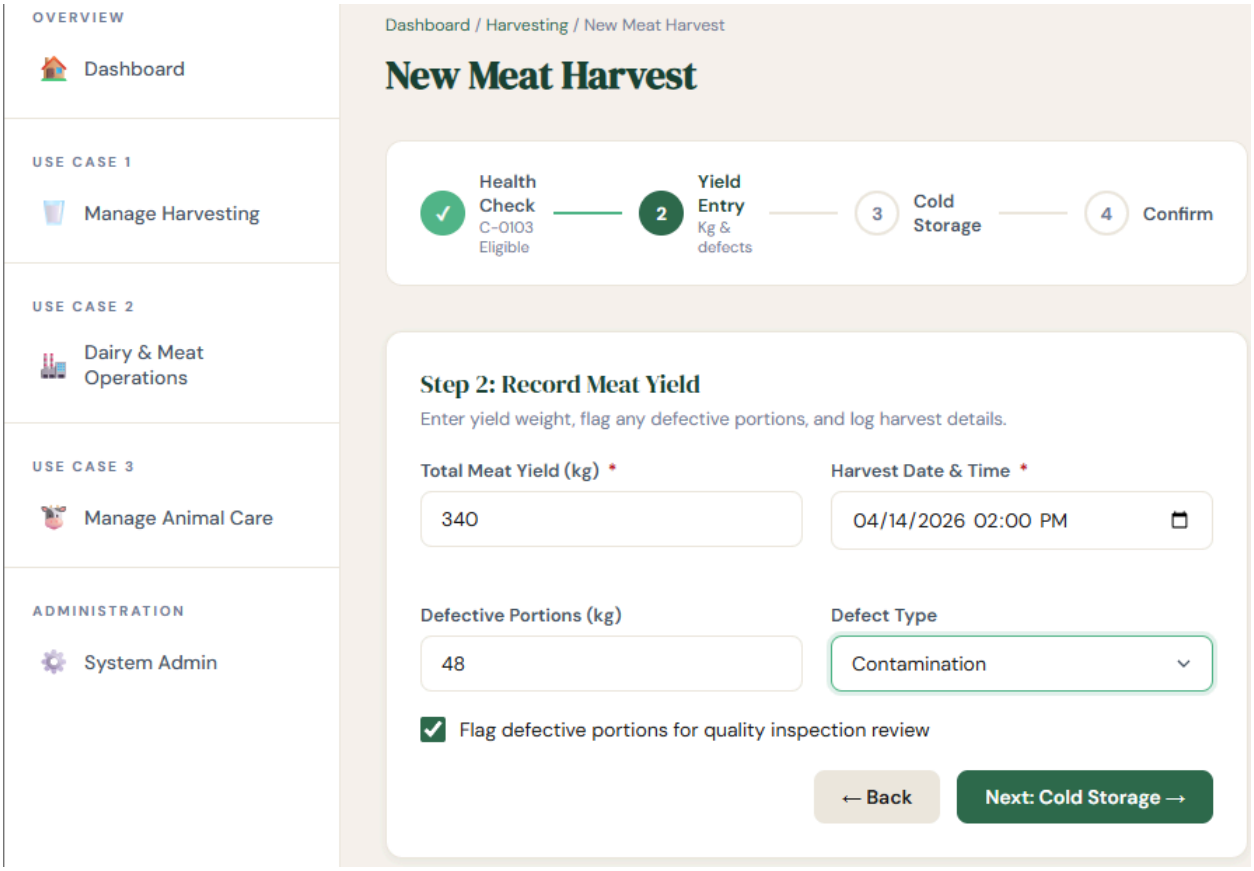
Step1: Retrieve Assessment Cow ID the farm worker needs to use Cow ID to retrieve assessment after that it has an information box of cow 1.COW-ID, 2. Type of raw material, 3. Weight, 4. Health status, 5. Vet Assessment, 6. Last check and click "Next: Yield Enter" to go to the next step.



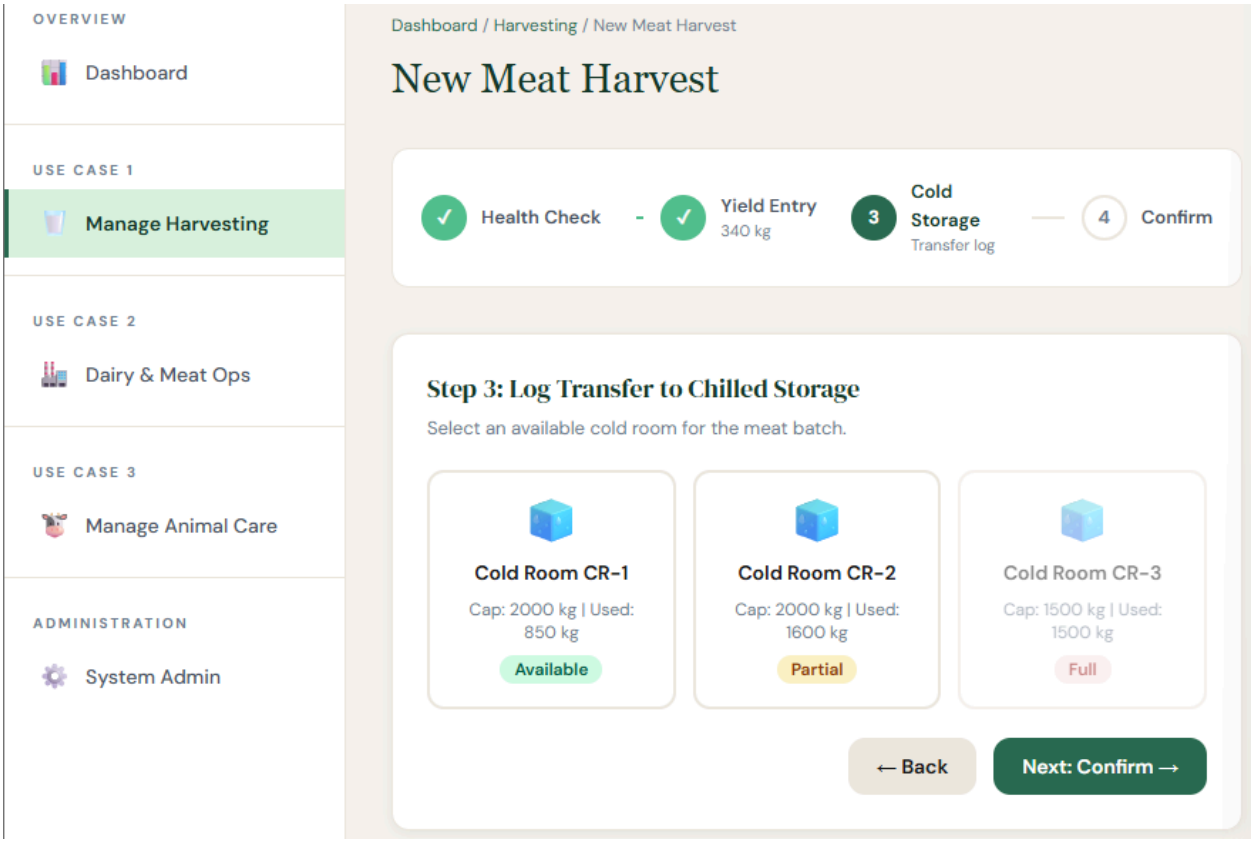
Step2: Record Meat Yield. This page allows employees to enter

1. Total Meat Yield (KG)
2. Harvest Date&Time
3. Defective Portions (KG)
4. Defect Type(Bruising, Contamination, Abnormal Colour and Other)

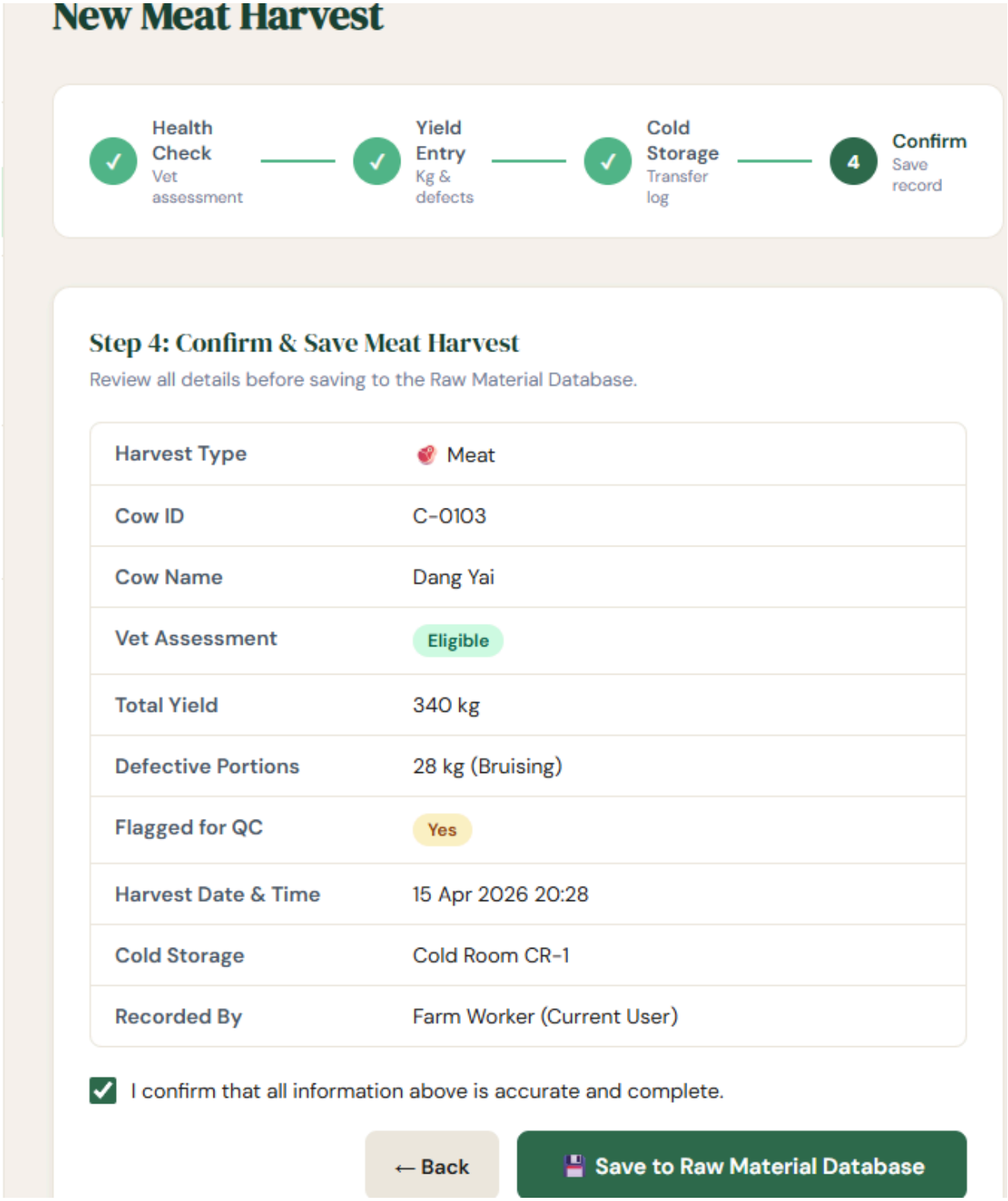
And click “Next: Cold Storage” to go to the next page.



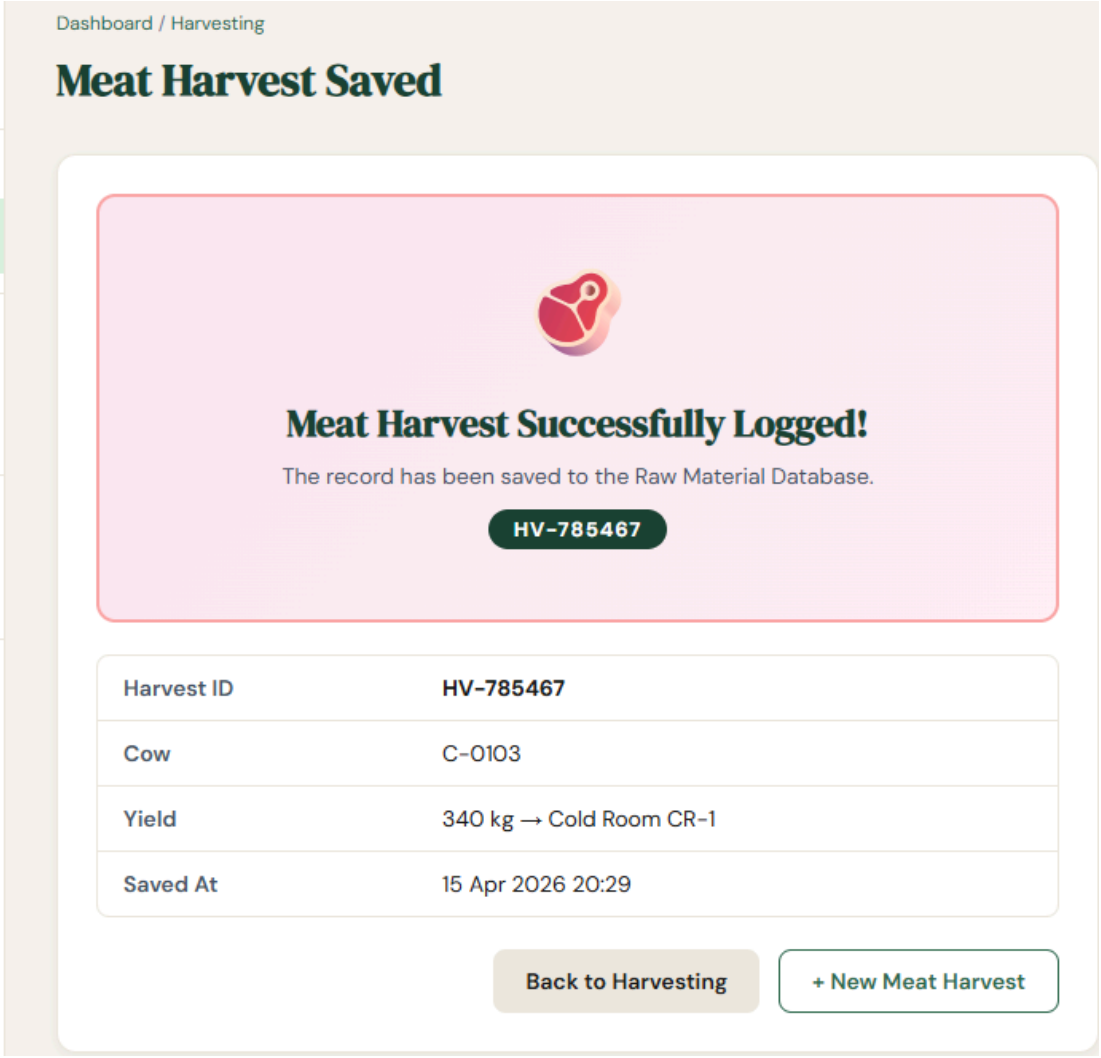
Step 3: Cold Storage and Log transfer to chilled storage. The farmworker must select the Cold Room. If a cold room still has space to hold raw material, it will be highlighted in green as “Available”. If a cold room is more than half full, it will be indicated in yellow as "Partial." If a cold room is completely full, it will be indicated in red as "Full" and can no longer be used. And for each room will display the current number of meat kg and full capacity.



Step 4: Confirm the information. This page displays all information farmworker fill to the system and after that farmworker need to click “Save to Raw Material Database” to save data.

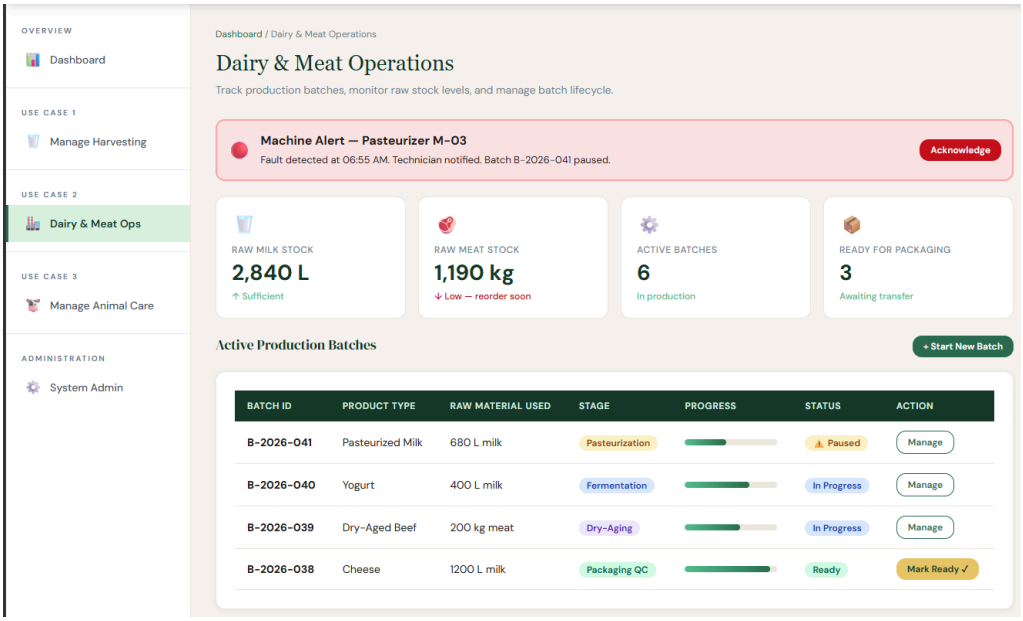


Finish meat harvest record saved page. It will show the Harvest ID, yield go to which cold room and Saves at date&time . And have a “Back to Harvesting” button and a “+New Meat Harvest” button for adding another new meat harvest..

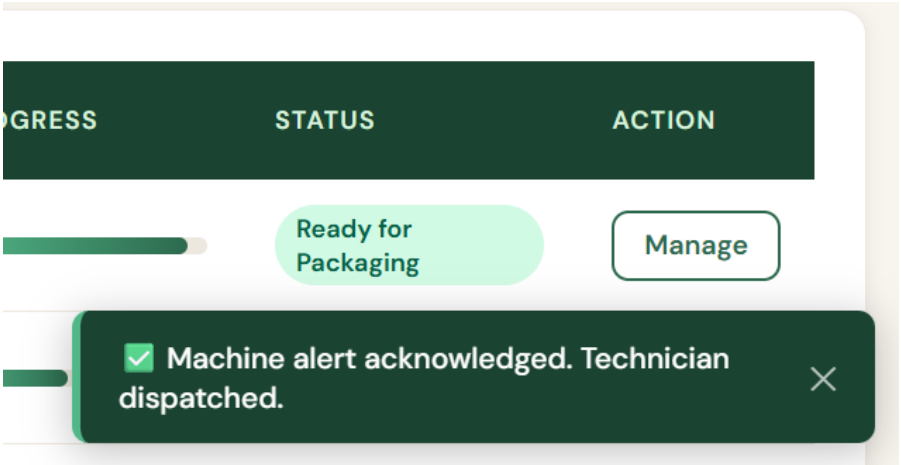


Dairy & Meat Operations

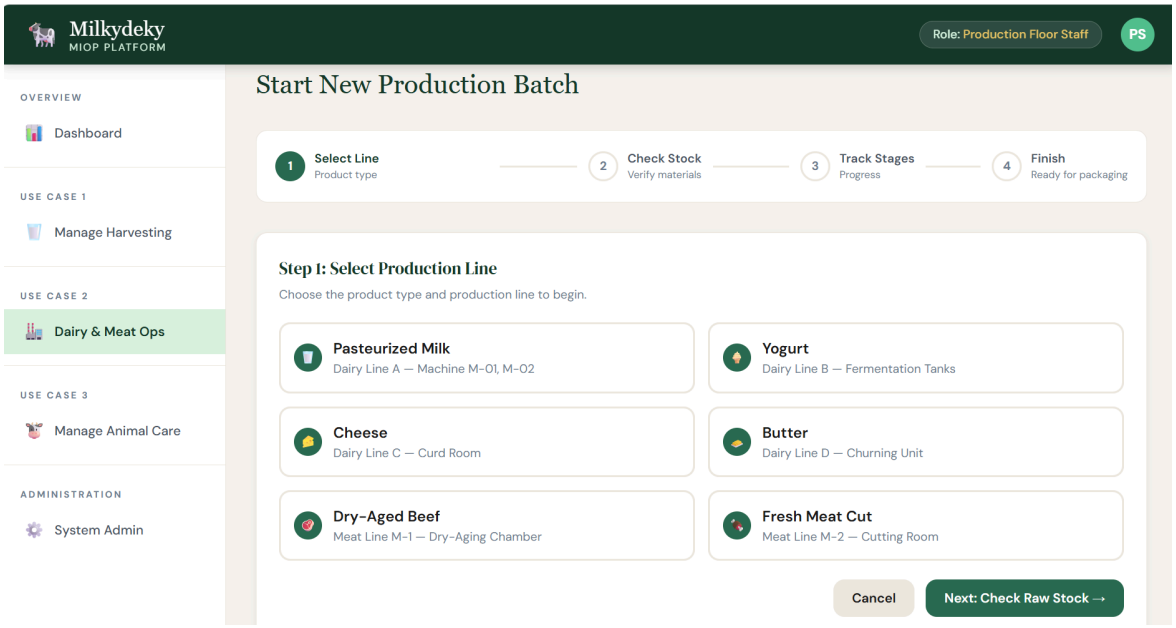
Description: This page is the “Dairy & Meat Operations” page, intended for Production Floor Staff to manage milk and meat stock data, including recording product information that needs to be delivered to customers. And the Production Floor Staff can check active production, which will record all information such as id, type of product, the amount of milk and beef used, as well as the status indicating the condition of that product. Additionally, it is also possible to check whether the machine is malfunctioning, as alerts will be displayed.



When clicking ‘Acknowledge’ to check the machine alert:

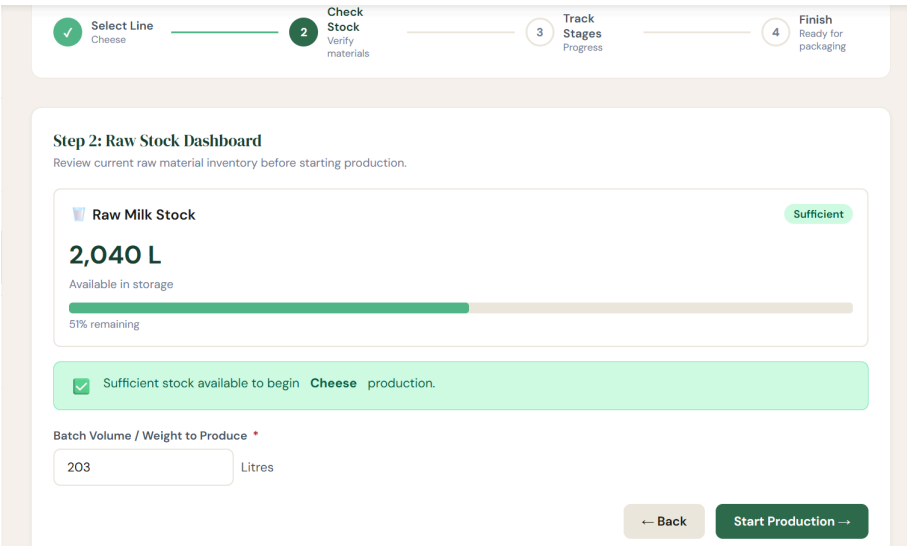


Step 1: When clicking the “Start New Batch” button. The system will navigate to the 'Start New Production Batch' page. In the first step, the Production Floor Staff must select the product type that they want to create (cheese yogurt butter etc.).

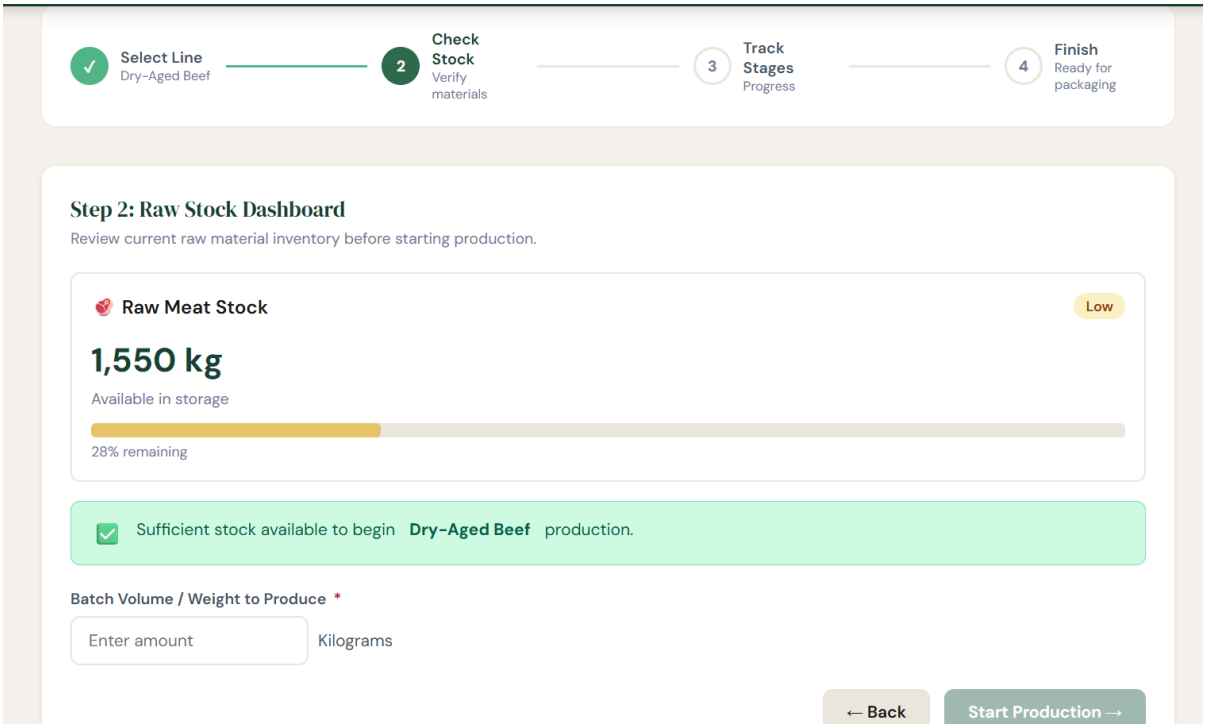


Step 2: When clicking the “Next: Check Raw Stock” button. The system will enter the second step, which is checking the raw material in stock and selecting the quantity needed. The system will automatically choose the raw materials required for production.

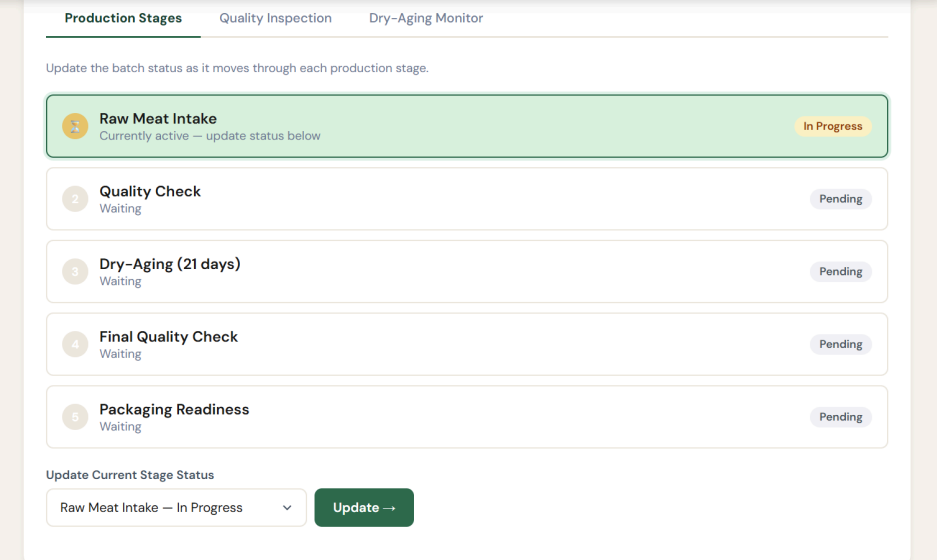
Step 2.1: The product that needs Raw Milk Stock.



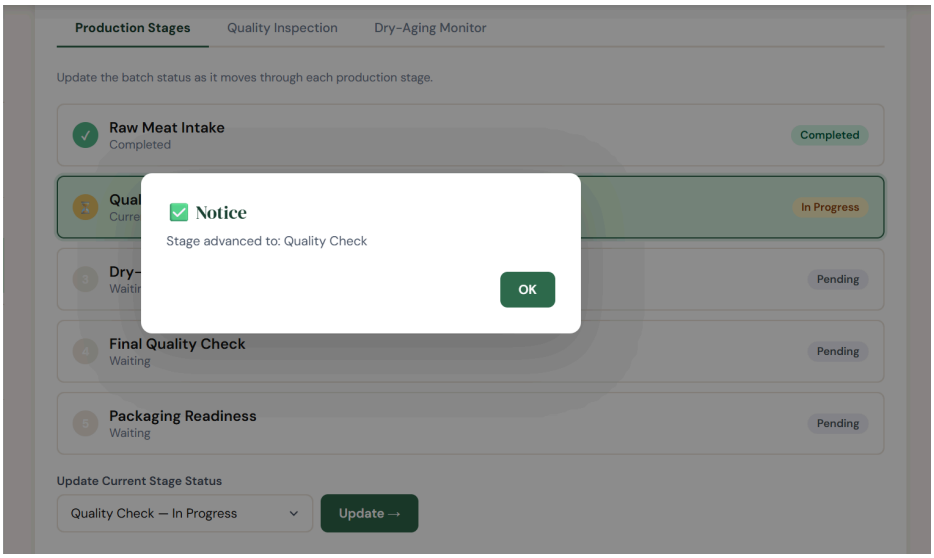
Step 2.2: The product that needs Raw Meat Stock.



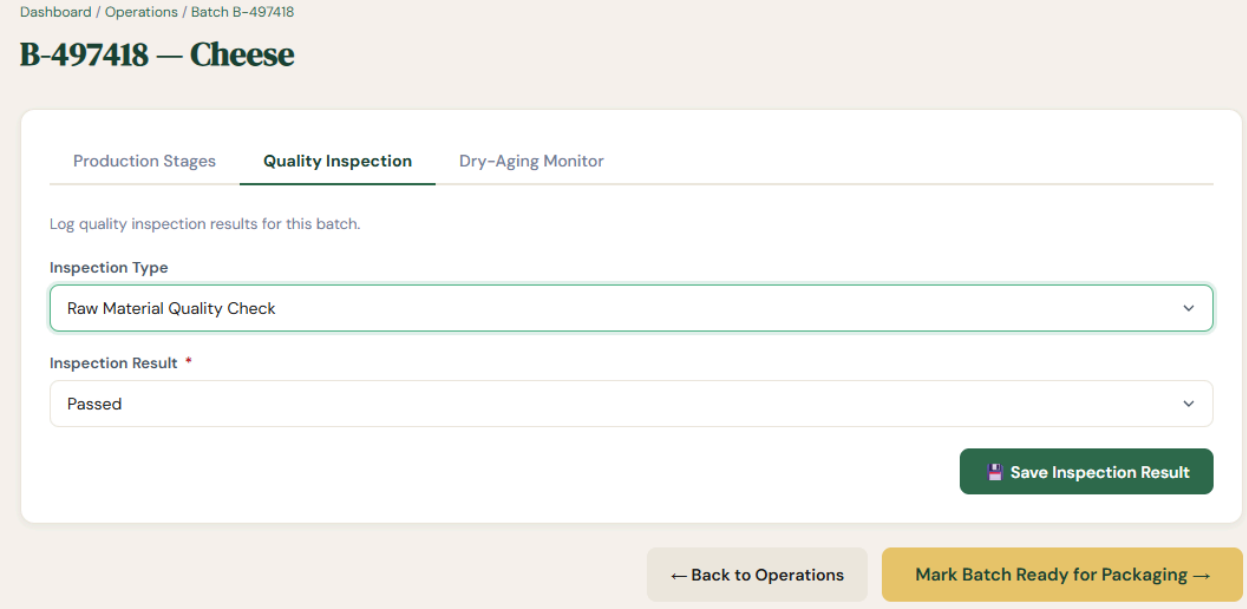
Step 3.1: When clicking the “Start Production” button. The system will show the production stage that is progressing and the Production Floor Staff can update the status at any time at 'Update Current Stage Status' by choosing between 2 options: In progress and Completed in every stage.



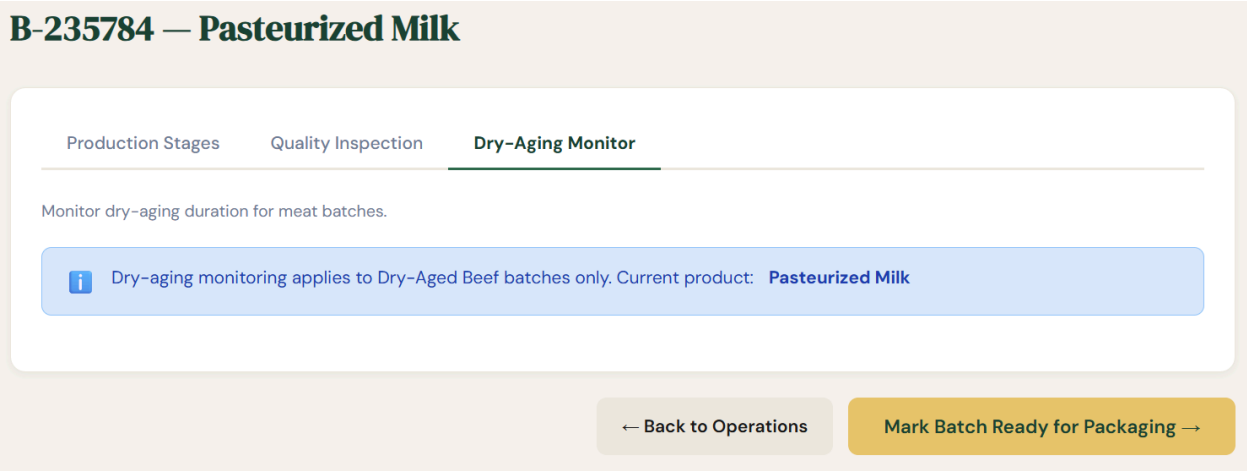
When click Update:



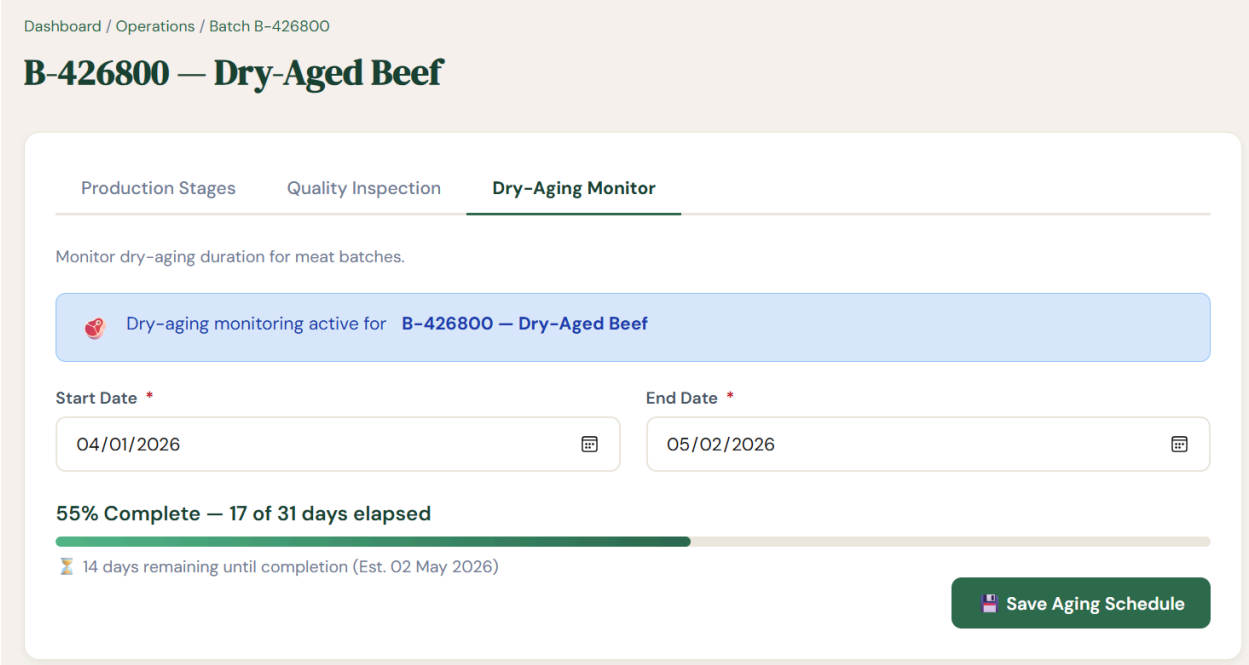
Step 3.2: In the “Quality Inspection” section, the system allows the Production Floor Staff to select the inspection type: Raw material quality check, Processing check, Packaging Readiness inspection. In addition, the Production Floor Staff also need to select the Inspection Type and Inspection Result such as Pass, Fail,Conditional.



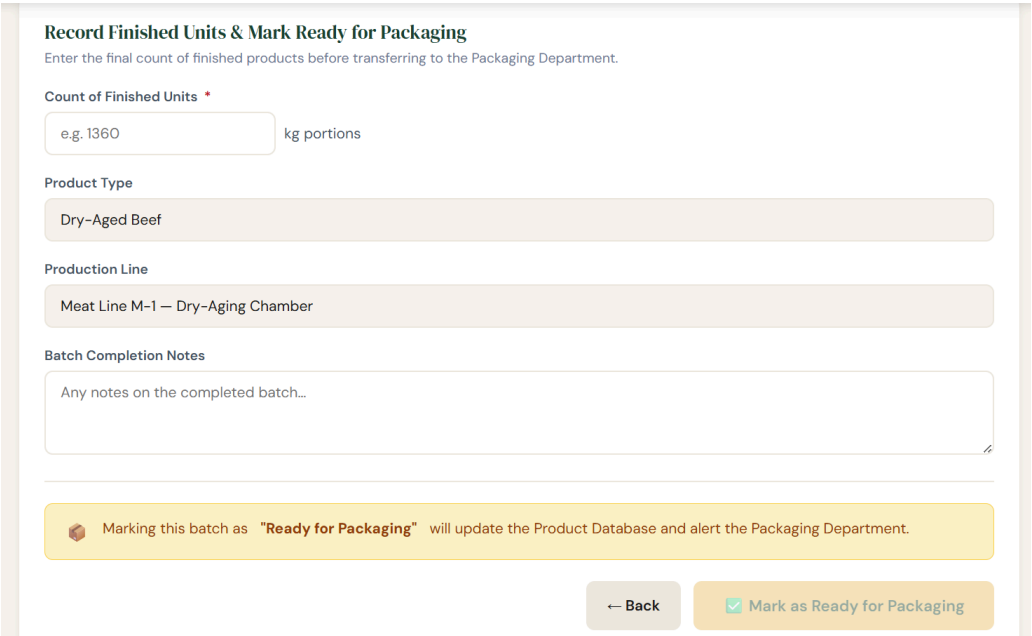
Step 3.3: As for the Dry-Aging Monitor, the system is limited to Dry-Aged Beef only (in this example, this product is making pasteurized milk, it cannot interact with this part).



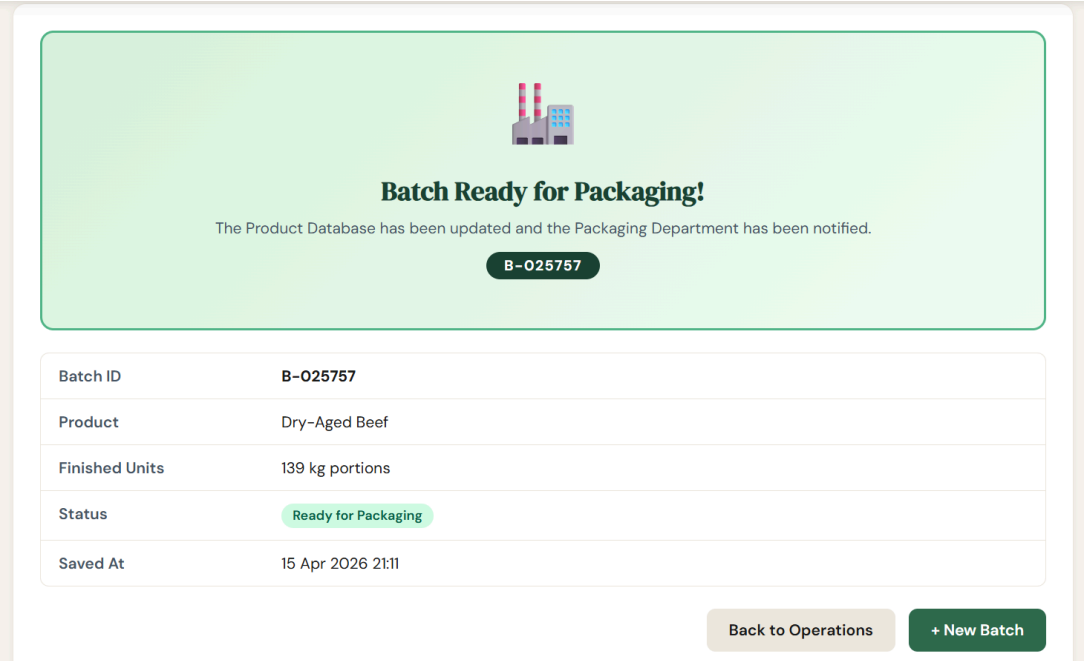
When selecting Dry-Aged Beef, the Production Floor Staff can select the Start Date and End Date process after pressing Save Aging Schedule. The user will be able to see the time progress.



Step 4: When clicking the "Mark Batch Ready for Packaging" button. The system allows the Production Floor Staff to select the count of finished units.

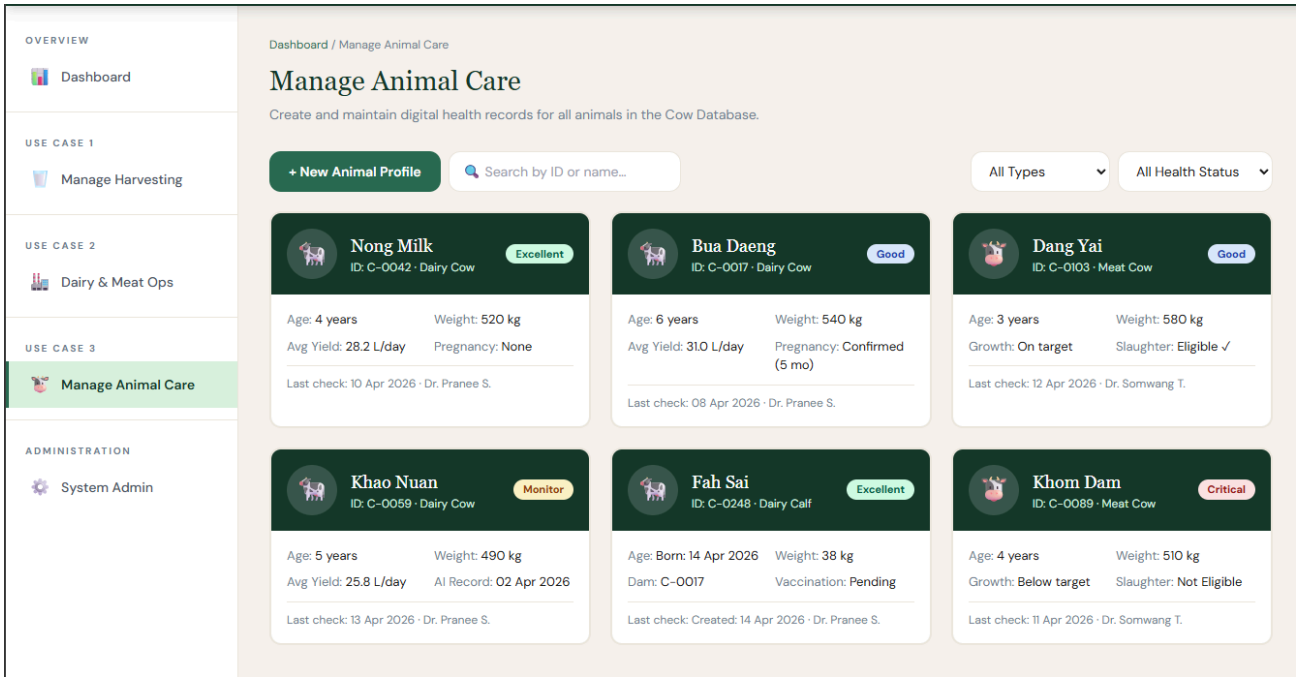


Step 5: Finally, The system will show that the batch is ready for packaging, along with various information and the date the data was recorded.

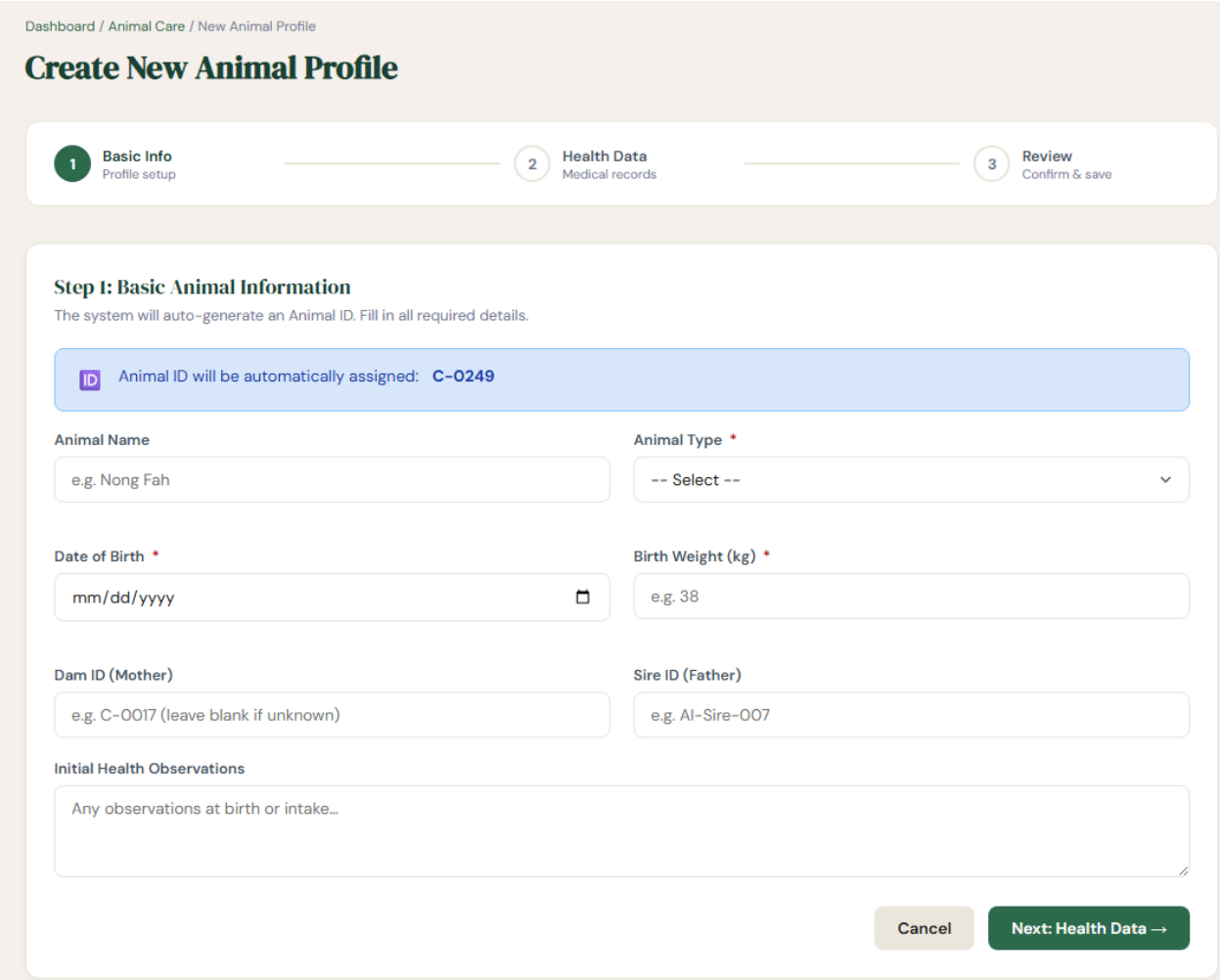


Manage Animal Care

Description: This is the “Manage Animal Care” page, which is designed for Veterinarians and Farm Workers to maintain animal health in the farm. The dashboard provides an overview of each animal's health status, examples like growth stats and pregnancy updates that centralize all health and reproductive history for the medical staff. Each cow has its own profile with color-coded status tags, so the on-duty veterinarian can identify which animals need a check up. It also tracks specific details like daily milk yield for dairy cows or slaughter readiness for meat cows, making sure every animal is healthy before moving to the next stage of production.

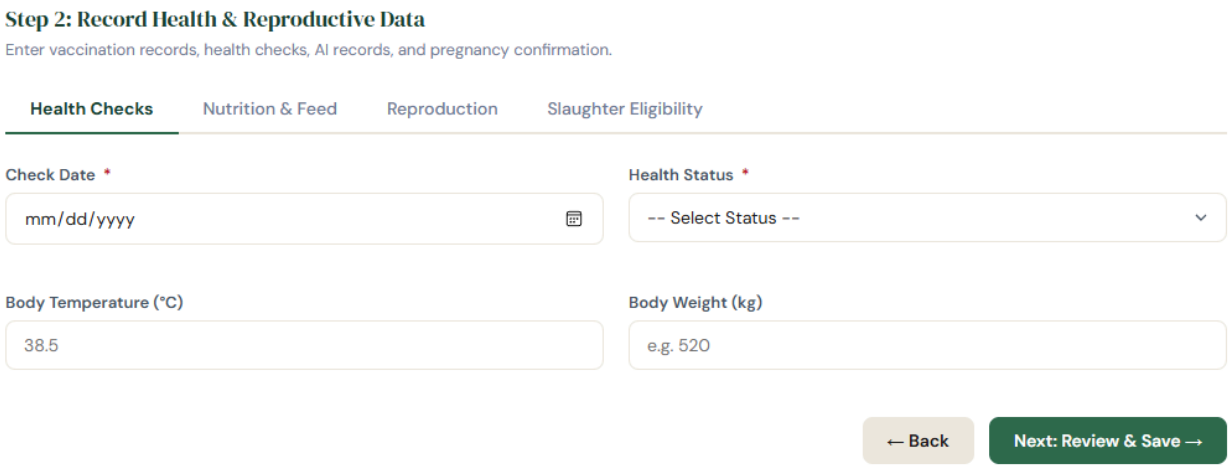


Step 1: This step is required to fill in an information form to create a new animal profile and also generate an animal ID upon completion of the form.



Step 2: This Step is required to fill in the health data of the animal, and can be branch into many sub-steps.

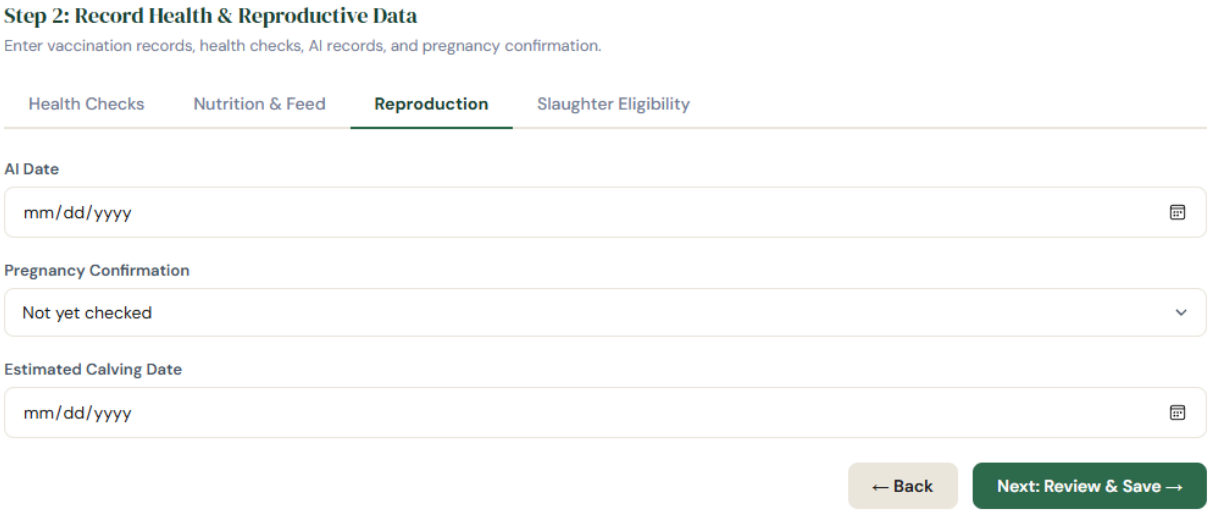
Step 2.1: In the image below shows you a form that required a health checks record of the animal.



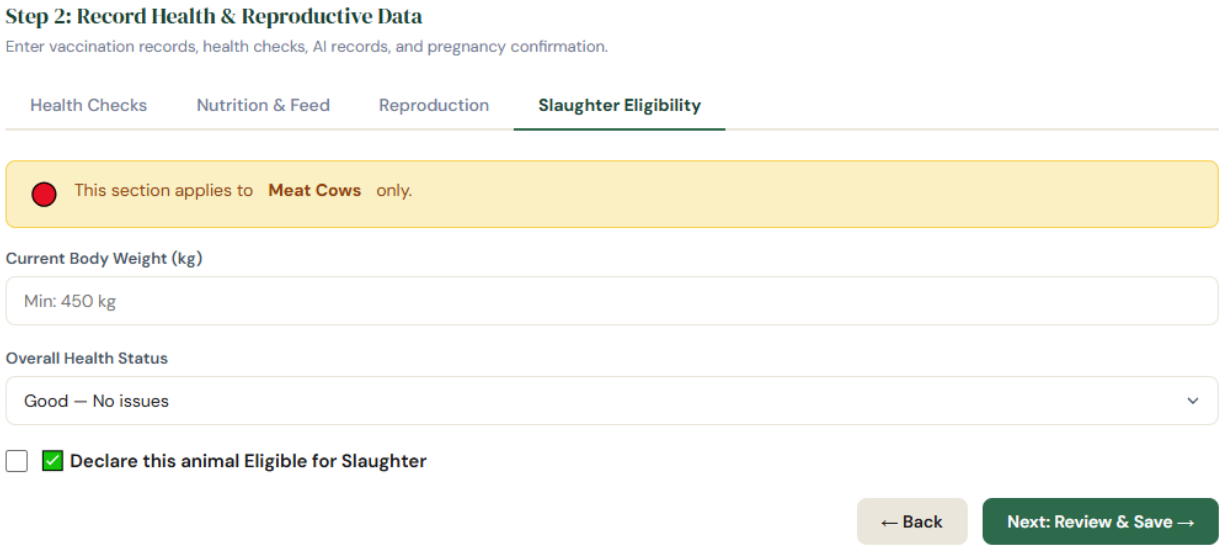
Step 2.3: The image below shows a form that requires recording an animal's nutrition details, including feed type, daily feed amount, nutrition status, growth milestones, and record date.



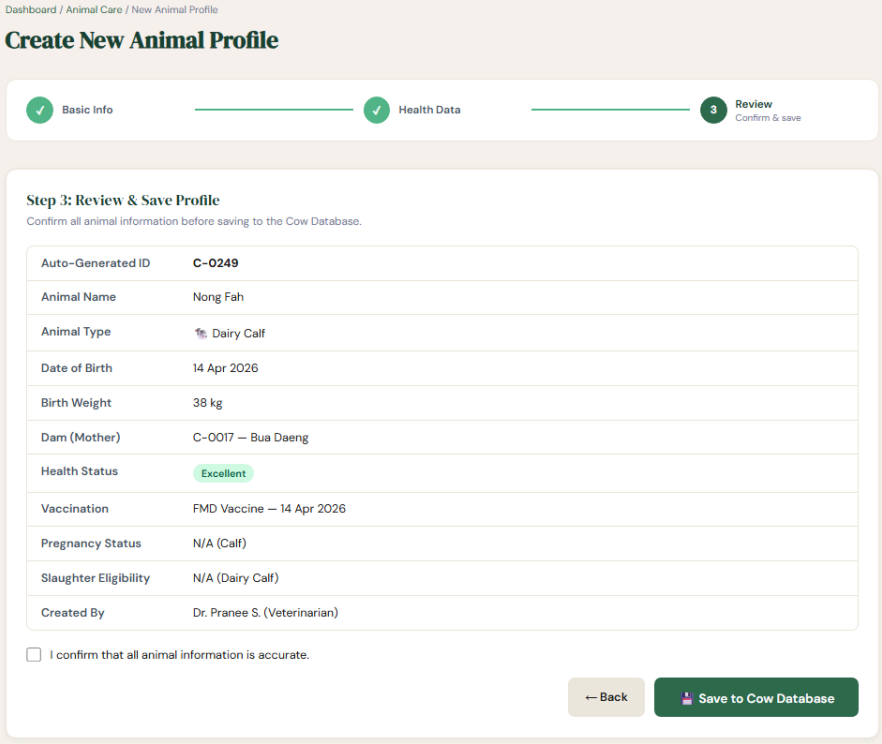
Step2.4: In the image below shows you a form that required a reproduction record of the animal.



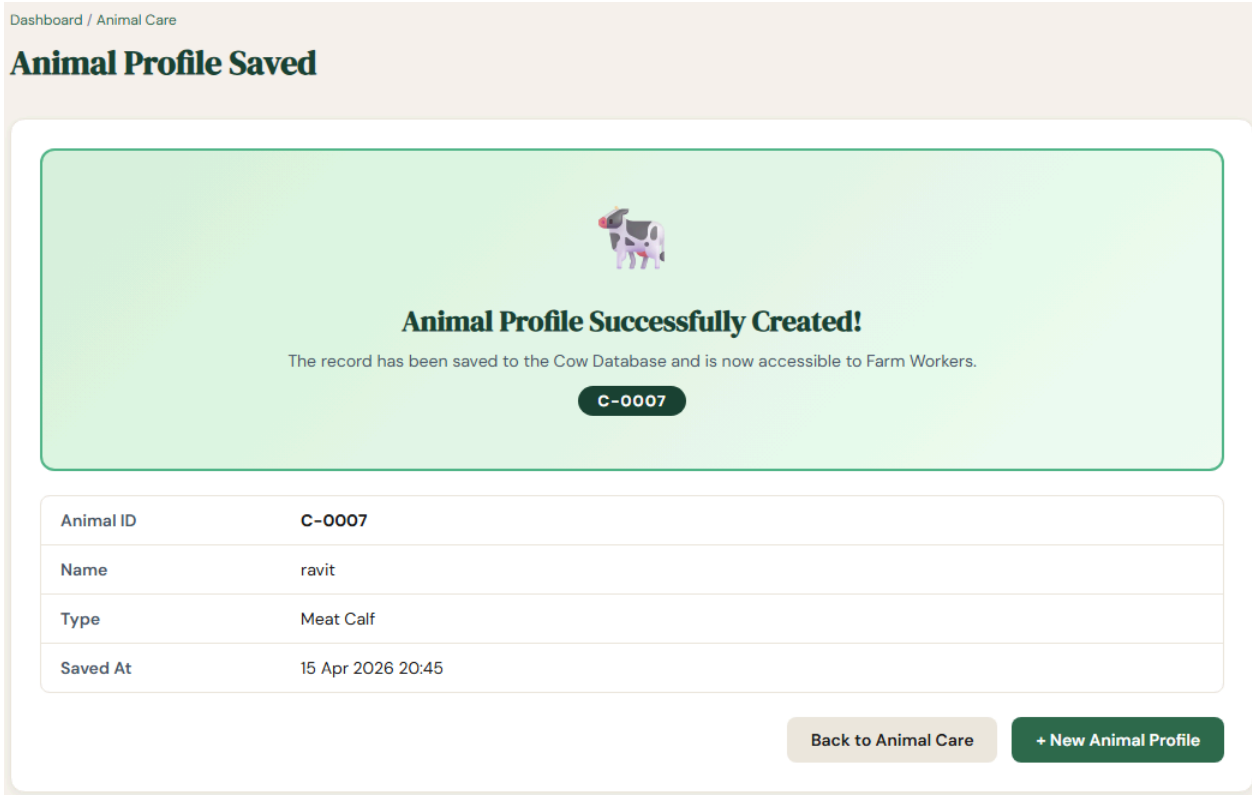
Step2.5: The image below shows you a form that required current body weights, overall health status, veterinarian assessment notes, and in this step the form allowed the declaration of slaughter eligibility of the animal.



Step3: This step allows veterinarians to review and confirm all animal information before saving to the Cow Database.



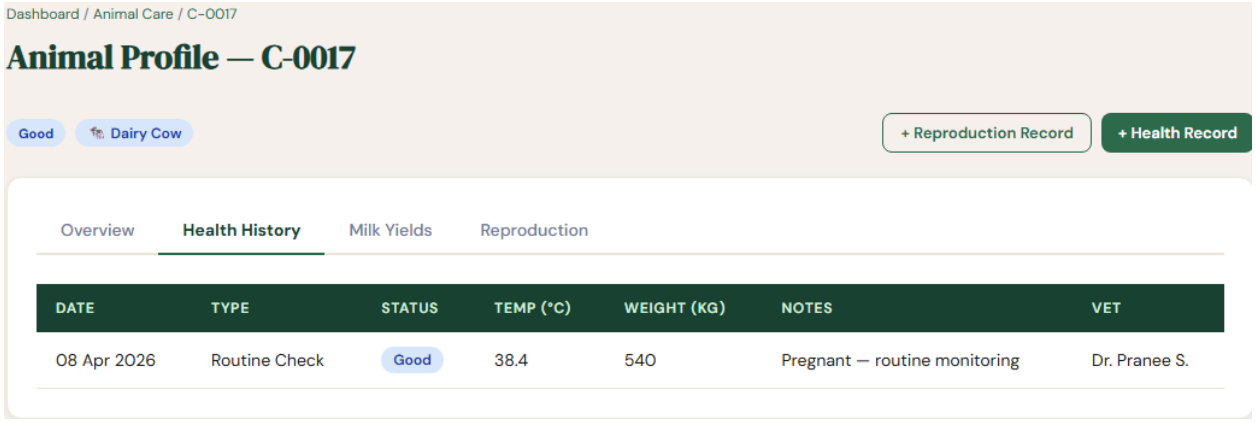
The image below shows that animal profile has been successfully created.



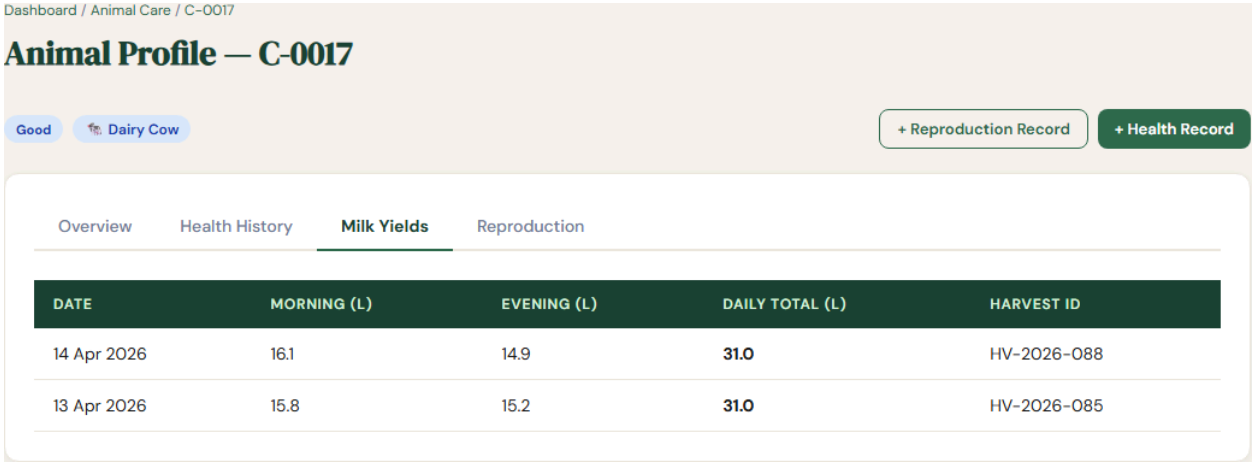
This image shows the animal profile of each animal and this tab is the overview tab to show all of the animal information.



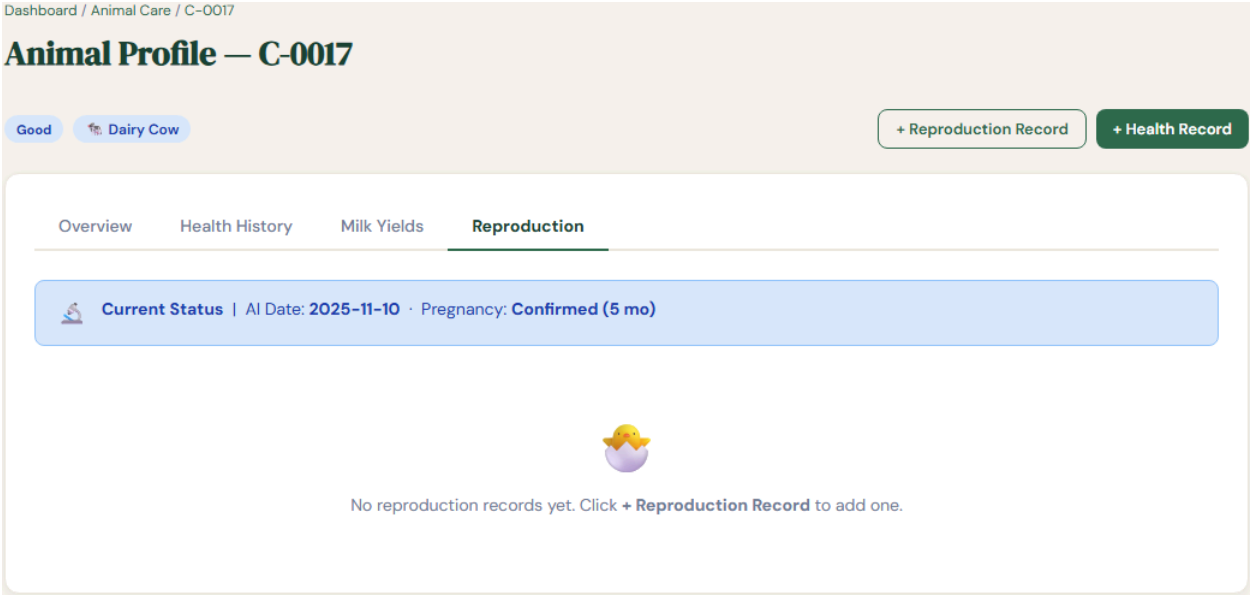
The image below shows the tab called health history that let the veterinarians assess the health history of the animal.



The image shows milk yields of the animal of each time including morning and evening and daily total in litres.



The reproduction tab on animal profile let the veterinarians assess the record of reproduction of the animal.



Test cases

Manage Harvesting Module (Meat)

Project Name: Milkydeky	
Test Case ID: TC-HV-001	Test Designed by: <Vachiravit Jantayasakorn>
Test Priority (Low/Medium/High): High	Test Designed date:<17/04/2026>
Module Name: **Manage Harvesting**	Test Designed by:<Vachiravit Jantayasakorn>
Test Title: To successfully add a meat record for harvesting. **(Happy path)**	Test Execution date:<17/04/2026>
Description: Verify that a farm worker can record a new meat harvesting with all required fields filled out correctly.	

Pre-conditions: User has valid Cow ID
Dependencies: Dependency on the Manage Animal Care module to have previously created the animal profile.

Step	Test Steps	Test Data	Expected Result	Actual Result	Status (Pass/Fail)	Notes
1	Go to manage harvesting page	-	Enter to the manage harvesting page	Enter to the manage harvesting page	Pass	
2	Click “+ New Milk Harvest”	-	Go to “Retrieve Health Assessment“	Go to “Retrieve Health Assessment“	Pass	
3	Provide Meat Cow ID	Cow ID = C-0103	Input cow ID successfully	Input cow ID successfully	Pass	
4	Click”Retrieve Assessment”	-	Show cow information and retrieve successfully	Show cow information and retrieve successfully	Pass	

5	Click “Next:Enter Yield”	-	Go to page step2: Record Meat Yield	Go to page step2: Record Meat Yield	Pass	
6	Provide Yield Weight	Cow weight = 340 kg	Finish add cow weight	Finish add cow weight and	Pass	
7	Select Date&Time	Harvest Date & time = 04/15/2026 10:23 PM	Finish determine date and time	Finish determine date and time	Pass	
8	Insert Defective Portions(kg)	Defective Portions = 49 kg	Finish defective of meat	Finish defective of meat	Pass	

9	Defect Type	Contamination	Finish select defect type	Finish select defect type	Pass	
10	Flag defective portions	Click check box	Finish check flag defective portions for quality inspection review	Finish check flag defective portions for quality inspection review	Pass	
11	Click “Next: Cold Storage”	-	Go to next page “Step3: Log Transfer to Chilled Storage”	Go to next page “Step 3: Log Transfer to Chilled Storage”	Pass	
12	Provide Storage Transfer (Cold Room)	Cold Room CR-1	Finish select cold room	Finish select cold room	Pass	

13	Click “Next: Confirm”	-	Go to next page “Step4: Confirm & Save Meat Harvest”	Go to next page “Step4: Confirm & Save Meat Harvest”	Pass	
14	Flag confirm all information	Click check box	Finish check flag confirm all information	Finish check flag confirm all information	Pass	
15	Check the information and click Save to Raw Material Database	-	Finish record new meat harvest	Finish record new meat harvest	Pass	
16	Click “Back to Harvesting” or “+New Meat Harvest”	-	Go back to Manage Harvesting page or back to “Step1 : Retrieve Health Assessment”	Go back to Manage Harvesting page or back to “Step1 : Retrieve Health Assessment”	Pass	

Manage Harvesting Module (Meat)

Project Name: Milkydeky	
Test Case ID: TC-HV-002	Test Designed by: <Chavanon>
Test Priority (Low/Medium/High): High	Test Designed date:<17/04/2026>
Module Name: **Manage Harvesting**	Test Designed by:<Chavanon>
Test Title: To alert message to the user when enter wrong Cow ID **(Unhappy path)**	Test Execution date:<17/04/2026>
Description: Farmworkers entered an incorrect Cow ID, and the system displays an alert.	

Pre-conditions: User has invalid Cow ID
Dependencies: Dependency on the Manage Animal Care module to have previously created the animal profile.

Step	Test Steps	Test Data	Expected Result	Actual Result	Status (Pass/Fail)	Notes
1	Go to manage harvesting page	-	Enter to the “manage harvesting” page	Enter to the manage harvesting page	Pass	
2	Press “+ New Meat Harvest"	-	Enter to the verify “Cow ID” page	Enter to the verify Cow ID page	Pass	
3	Provide Meat Cow ID	Cow ID = C-0000	Input successfully	Input successfully	Pass	
4	Click “Retrieve Assessment”	-	Show message alert to user”Cow ID not found or not a Dairy Cow. Please check the ID.”	Show message alert to user”Cow ID not found or not a Dairy Cow. Please check the ID.”	Pass	

Dairy & Meat Operations

Project Name: Milkydeky	
Test Case ID: TC-D&M-003	Test Designed by: <Chavanon>
Test Priority (Low/Medium/High): High	Test Designed date:<17/04/2026>
Module Name: **Manage Dairy & Meat Operations**	Test Designed by:<Chavanon>
Test Title: Start new production batch successfully**(Happy path)**	Test Execution date:<17/04/2026>
Description: Production floor worker create new production batch	

Pre-conditions: User has valid Sufficient Raw material
Dependencies: Dependency on the Manage Harvesting module to have sufficient stock available for production.

Step	Test Steps	Test Data	Expected Result	Actual Result	Status (Pass/Fail)	Notes
1	Go to Dairy & Meat Operations	-	The page load successfully	The page load successfully	Pass	
2	Press “Start New Batch” Button	-	Load to “select production line” page	Load to “select production line” page	Pass	
3	Select Product Type	Cheese	Product type selection successful	Product type selection successful	Pass	
4	Press “Next: Check Raw Stock”	-	Show the “Raw Stock Dashboard" page	Show the “Raw Stock Dashboard" page	Pass	

5	Select Material Used	500 L milk	System call accept the value	System call accept the value	Pass	
6	Press “Start Production”	-	Load to “Raw Stock Dashboard" page	Load to “Raw Stock Dashboard" page	Pass	
7	Select to “Update Current Stage Status”	Update Current Stage Status = “Update Raw Milk intake process to be completed”	Show “Stage advanced to: Next Stage”	Show updated successfully	Pass	

8	Click “Update”	-	Status update to completed	Status update to completed	Pass	
9	Click “Quality Inspection”	-	Go to next part Quality Inspection	Go to next part Quality Inspection	Pass	
10	Insert information to Quality Inspection	Inspection Type = “Raw Material Quality Check” Inspection Result = “Passed”	Input Quality Inspection successfully	Input Quality Inspection successfully	Pass	
11	Click “Save Inspection Result”	-	Save Inspection Result successfully	Save Inspection Result successfully	Pass	
12	Press ”Mark Batch Ready for Packaging”	-	Load to “Finalize Batch” page	Load to “Finalize Batch” page	Pass	

13	Insert Count of Finished Units	10 cartons / packs	System call accept the value	System call accept the value	Pass	
14	Press ”Mark as Ready for Packaging”	-	Show Batch Completed	Show Batch Completed	Pass	
15	Press“Back to Operation” or “+New Batch”	-	Finish batch for packaging and back to “Dairy&Meat Operation” or “Start New Production Batch”	Finish batch for packaging and back to “Dairy&Meat Operation” or “Start New Production Batch”	Pass	

Dairy & Meat Operations

Project Name: Milkydeky	
Test Case ID: TC-D&M-004	Test Designed by: <Vachiravit>
Test Priority (Low/Medium/High): High	Test Designed date:<17/04/2026>
Module Name: **Manage Dairy & Meat Operations**	Test Designed by:<Vachiravit>
Test Title: Alert message **(Unhappy path)**	Test Execution date:<17/04/2026>
Description: Alert message when you don't have input on “Raw Stock Dashboard” page that needs to input “Batch Volume” in Litres.	

Pre-conditions: Empty
Dependencies: Dependency on the Manage Harvesting module to have sufficient stock available for production.

Step	Test Steps	Test Data	Expected Result	Actual Result	Status (Pass/Fail)	Notes
1	Go to Dairy&Meat Operations page	-	Enter to Dairy&Meat Operations page	Enter to Dairy&Meat Operations page	Pass	
2	Click “+Start New Batch”	-	Go to next page “Start New Production Batch”	Go to next page “Start New Production Batch”	Pass	
3	Select option on Production Line	Select Cheese	Finish selected production line	Finish selected production line	Pass	
4	Click “Next: Check Raw Stock”	-	Go to next page “Raw Stock Dashboard”	Go to next page “Raw Stock Dashboard”	Pass	

5	Click “Start Production”	empty	Alert message “ Please enter a batch volume before starting production.”	Alert message “ Please enter a batch volume before starting production. ”	Pass	

Manage Animal Care

Project Name: Milkydeky	
Test Case ID: TC-MAC-005	Test Designed by: <Phon>
Test Priority (Low/Medium/High): High	Test Designed date:<17/04/2026>
Module Name: **Manage Animal Care**	Test Designed by:<Phon>
Test Title: Create a new animal profile successfully. **(Happy path)**	Test Execution date:<17/04/2026>
Description: Verify that a farmworker can create a new animal profile with all required fields filled out correctly.	

Pre-conditions: User has valid Animal Type, Date of Birth and Birth Weight(kg)
Dependencies: None

Step	Test Steps	Test Data	Expected Result	Actual Result	Status (Pass/Fail)	Notes
1	Go to Manage Anime Care	-	Enter to “Manage Anime Care” page	Enter to “Manage Anime Care” page	Pass	
2	Press “New Animal Profile ”	-	Load to “Basic Animal Information” page	Load to “Basic Animal Information” page	Pass	
3	Insert Animal Name	Pai	System call accept the value	System call accept the value	Pass	
4	Select Animal Type	Dairy Cow	Finish selected Animal Type	Finish selected Animal Type	Pass	
5	Select Date of Birth	17/04/2026	Finish selected Date of Birth	Finish selected Date of Birth	Pass	

6	Insert Birth Weight (kg)	39	System call accept the value	System call accept the value	Pass	
7	Press “Next: Health Data”	-	Load to “Record Health & Reproduction Data” page	Load to “Record Health & Reproduction Data” page	Pass	
8	Insert Check Date into Health Checks	Check Date = 04/17/2026	Input “Check Date” successfully	Input “Check Date” successfully	Pass	
9	Choose Health status	Execellent	Finish selected Health Status	Finish selected Health Status	Pass	
10	Insert Body Temperature(°C)	Body Temperature(°C) = 38.5	Finish input “temperature” successfully	Finish input “temperature” successfully	Pass	
11	Insert Body Weight(kg)	Body Weight(kg) = 520	Finish input “Body Weight(kg)” successfully	Finish input “Body Weight(kg)” successfully	Pass	

12	Press “Next: Review & Save ”	-	Load to “Review & Save Profile” page	Load to “Review & Save Profile” page	Pass	
13	Flag “I confirm that all animal information is accurate. ” Button	Click checkbox	Checkbox “confirm that all animal information is accurate.” has been checked	Checkbox “confirm that all animal information is accurate.” has been checked	Pass	
14	Press “Save to Cow Database” Button	-	The cow information profile is ready to be verified.	The cow information profile is ready to be verified.	Pass	
15	Click “Back to Animal Care” or “+ New Animal Profile”	-	Back to Manage Animal Care page and create a new animal profile.	Back to Manage Animal Care page and create a new animal profile.	Pass	

Manage Animal Care

Project Name: Milkydeky	
Test Case ID: TC-MAC-006	Test Designed by: <Chavanon>
Test Priority (Low/Medium/High): High	Test Designed date:<17/04/2026>
Module Name: **Manage Animal Care**	Test Designed by:<Chavanon>
Test Title: Alert message **(Unhappy path)**	Test Execution date:<17/04/2026>
Description: Alert message when you don't have input on “Basic Animal Information” page that needs to input 1.Animal Type 2.Date of Birth 3.Birth Weight(kg).	

Pre-conditions: Empty
Dependencies: None

Step	Test Steps	Test Data	Expected Result	Actual Result	Status (Pass/Fail)	Notes
1	Go to Manage Anime Care	-	Enter to “Manage Anime Care” page	Enter to “Manage Anime Care” page	Pass	
2	Press “New Animal Profile ”	-	Load to “Basic Animal Information” page	Load to “Basic Animal Information” page	Pass	
3	Press “Next: Health Data”	-	Alert message Animal Type = “Please select an animal type. ” Date of Birth = “Please enter the date of birth. ” Birth Weight (kg) = “Please enter a valid birth weight (greater than 0). ”	Alert message Animal Type = “Please select an animal type. ” Date of Birth = “Please enter the date of birth. ” Birth Weight (kg) = “Please enter a valid birth weight (greater than 0). ”	Pass	Required inputs were never filled in.